

Title

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Abstract—Abstract...

Index Terms—molecular communication, agriculture, pest control.

I. INTRODUCTION

Introduce
Motivate
Literature Review
List Contributions (bullet points)
Outline of the paper

II. INTRACELLULAR SIGNALLING MODEL

This section specifies the reaction network of a single *Dicotyle discoideum* cell that is embedded in the reaction–diffusion simulation of the cell population. For every module we list the reactions and constants in tables, explain why each reaction is present and which experiments motivate it, justify the constants, and justify every Hill gate. The numbering of the modules is that of the simulation code; Modules 5–7 (Rac/PAK, PLA2/Ca²⁺ and Ca²⁺ influx) are specified in the model description but are not part of the network analysed here.

A. Overview and conventions

a) Signal flow.: Extracellular cAMP is sensed by the receptor cAR1 (Module 1), which catalyses the G-protein cycle (Module 2). Free Gβγ drives two Ras branches: RasG, whose adaptation and amplification (Module 3) control PI3K and PIP3 (Module 4), and RasC, which acts through TORC2, the kinases PKB and the effector CRAC (Module 8). Module 8 injects an activator into an excitable element (Module 0) that opens the adenylyl cyclase ACA (Modules 0 and 9) and thereby produces the cAMP pulse. Intracellular cAMP is exported (Module 9), degraded intracellularly and feeds back through PKA, ERK2 and RegA (Module 10), and is degraded extracellularly by the secreted phosphodiesterase PdsA and its inhibitor PdiA (Module 11), which closes the loop.

b) Spatial discretisation and stochastic engine.: The domain is a square lattice with voxel edge 2 μm, so that one voxel has a volume of 8 μm³ and 1 μM corresponds to $M = 4817.6$ molecules per voxel. A cell occupies $N_{\text{vox}} = 13$ voxels (a 3 × 3 block plus one voxel on each side), i.e. 104 μm³. In every time step of $\Delta t = 2$ ms, each voxel runs an

exact stochastic simulation of its reactions up to Δt [?], after which molecules move to neighbouring voxels; the diffusion step requires $D\Delta t/\Delta x^2 \leq 0.25$, which holds for the fastest species, cAMP ($D = 350 \mu\text{m}^2/\text{s}$, value 0.175). Membrane-bound and cytosolic species of the cell are confined to its voxels, whereas extracellular species (cAMP_e, PdsA_e, PdiA and their complex) live on the world lattice, which they leave at the domain boundary. Because the cost of a step scales with the total propensity, uniform rate multipliers change the run time.

c) Units and rate constants.: First-order constants are given in s^{−1}, bimolecular constants in μM^{−1}s^{−1} (the engine divides the propensity by M once), and zeroth-order sources in μMs^{−1} per voxel. Rate constants are effective values of the model, not always the microscopic constants of a single enzyme; where a value is a lumped or rescaled quantity this is stated.

d) Two numerical conventions that affect propensities.: The engine evaluates the raw product of the educt counts, so that 2X would react with itself at rate $\propto n^2$ instead of $n(n-1)$. At the copy numbers of the excitable element (a few molecules per cell at rest) this is a 23 % error, and it acts as a spurious first-order source when molecules are spread over 13 voxels. Same-species reactions are therefore written with one educt and a falling-factorial factor (Table ??), which makes the per-voxel decomposition exact and the 13-voxel cell equivalent to one well-mixed compartment. Second, Hill gates with $n \geq 6$ are written in the reciprocal form $[1 + (K/x)^n]^{-1}$, because the propensity arithmetic is single precision and x^8 overflows for $x > 65\,536$, which would make the gate undefined.

e) Provenance of the constants.: Every constant carries a basis tag in the constants tables: **L** a literature value used directly; **D** derived from a literature anchor by a stated relation; **C** calibrated on the model against a stated target behaviour, which is not an independent measurement; **E** an estimate or design choice for which no measurement is available. Tags C and E are the ones a reader should treat with caution. The number of C/E constants is large by necessity: many molecular abundances and rates of this pathway have never been measured, and the calibrations aim at reproducing the measured *behaviour* (kinetics, thresholds, periods) of the cell rather than individual molecular values.

f) Constants that are deliberately not literature values.: Three departures are made explicit because they trade realism for signal-to-noise or tractability: (i) the G-protein to receptor ratio is 60 instead of about 3 (Module 2) to raise molecule counts in the gradient-sensing stages; (ii) the uniform acceleration of all Module-4 rates by a factor of 20, which compresses time scales that were not measured; (iii) the PdsA–PdiA dissociation constant is 10⁴ times weaker than measured (Module 11).

This paper was presented in part at the 13th ACM International Conference on Nanoscale Computing and Communication (cite the confi here) [DOI: add doi later].

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TABLE I
HILL GATES AND SATURABLE KINETICS OF THE DEFAULT NETWORK. x IS THE INPUT OF THE GATE.

Gate	Reaction	Form	n	K	Justification
Competence $c(H)$	0.9a	$H^n/(K^n + H^n)$	4	60 molec/voxel	Developmental switch between non-firing and firing cells; $c(100) = 0.885$. Phenomenological (Hunger is an abstract clock), steep to give a sharp onset.
Refractory block $b(Rf)$	0.9a, 8.14	$K^n/(K^n + x^n)$	4	450 molec/cell	Decreasing gate that must shut the seed and the relay hard after a pulse and reopen within minutes. K and τ_R were set on the measured recovery curve (blocked at 300 s, open by 420 s); $n = 4$ makes the boundary sharp.
ACA gate	0.9f	$x^n/(K^n + x^n)$	4	192 molec/voxel ERK2*	Opens the cyclase only after ERK2* has disarmed RegA (ERK2 is needed for ACA activation [?]) and controls the phosphodiesterase RegA [?], [?]); enforces the order ERK2→ACA.
PTEN eviction	4.6c	$x^n/(K^n + x^n)$	2	1.31×10^5 PIP3/voxel	Realises mutual inhibition PIP3 $\dashv$ PTEN [?]; $n = 2$ is the minimum cooperativity for winner-take-all behaviour. K (17 % of the lipid pool) sits on the steep flank; beyond ≈ 19.5 % the domain becomes an orientation-independent latch.
PIP3 relay gate g	8.14	$[1 + (K/x)^n]^{-1}$	8	5000 PIP3/voxel	The relay must open at the PIP3 evoked by 2 nM cAMP (≈ 5000 /voxel; gate 0.51) and stay shut at the resting level ($\approx 10^3$; gate $< 10^{-5}$), a contrast of only about 5. The exponent $n = 8$ was set when the peak-to-plateau ratio was 2.7, for which a 300-fold suppression of the standing drive needs $n \approx 8$, and was kept when K was re-anchored. It stands for the lumped chain PIP3→PKB→CRAC→cyclase priming, not one binding event. Reciprocal form (Sec. ??).
NF1 saturable GAP	3.7z	$x/(K + x)$	1	$0.2 \mu\text{M}$	Michaelis–Menten hydrolysis with $K \ll [\text{Ras}]_{\text{tot}}$ (zero-order regime), which produces rectification of Ras activation [?]; K from [?], [?].
PdsA induction $P(H)$	11.D1b, 11.E1	$H^n/(K^n + H^n)$	2	25 molec/voxel	Starvation gate of the aggregation-stage <i>pdsA/pdiA</i> expression; opens early relative to the Hunger ramp, $P(100) = 0.94$.
PdiA repression r	11.E1	$[1 + (x/K)^n]^{-1}$	2	$0.05 \mu\text{M}$ cAMP _s	Repression of PdiA synthesis by the slowly filtered cAMP signal [?]; $n = 2$ gives a graded but steeper-than-hyperbolic switch.
Memory writing (optional)	M.1	$x^n/(K^n + x^n)$	3	5000 PIP3/voxel	Writes the mark only on the PIP3 rise ($H(\text{rest}) \approx 0.03$, $H(7500) \approx 0.77$).

B. Hill gates

Hill functions $x^n/(K^n + x^n)$ appear in the network whenever a lumped, thresholded step replaces a multi-step process that is not resolved by the model. They are not used to claim molecular cooperativity of n binding events. A gate is justified when (i) a switch-like transition is required for the function of the module, (ii) its half-point K can be placed relative to the resting and stimulated values of its input, and (iii) its steepness n is the smallest value that achieves the required contrast. Table ?? lists all gates of the default network, with the value of the gate at rest and at the stimulated level of its input where these are known from the model. Two further rules were enforced when placing the half-points: the activated fraction of a switch must be neither near zero nor near one at its working point (a saturated gate cannot report a difference), and each K is placed between the resting and the driven value of its input.

C. Reproducibility

All tables below are generated from the values that the simulation loads by default; every constant can be overridden from

the environment for sensitivity studies. Two configurations of the code (a two-state and a multi-affinity receptor, Module 1) exist; only the two-state receptor is used.

D. Module 0: Developmental State, Excitable Element and Refractory Clock

a) Role.: Before collective waves start, isolated starved cells emit cAMP pulses spontaneously, on average about once per 12 min and at most about once per 4 min, with inter-pulse intervals that follow a gamma distribution (shape 6.6, rate 1.25 min^{-1}) [?]. Such noise-driven pulsing of single cells below the threshold for collective behaviour is a key step in the onset of oscillations [?], [?]. The molecular circuit that generates periodic cAMP in populations couples cAMP-induced activation of ACA and ERK2, inhibition of ERK2 by PKA and phosphorylation of the phosphodiesterase RegA by ERK2 [?], [?], and produces a period of about 7 min. Module 0 replaces the unresolved part of this circuit by a compact excitable element and adds the two variables that the collective behaviour needs: a developmental state and a refractory clock.

TABLE II

MODULE 0 REACTIONS: HUNGER, THE EXCITABLE ELEMENT (X, Y), THE REFRACTORY CLOCK (Rf) AND THE ERK2→ACA OUTPUT STAGE. FOR THE SPECIES OF THE EXCITABLE ELEMENT N_s IS THE COPY NUMBER OF SPECIES s per cell; $c(H)$ AND $b(Rf)$ ARE THE HILL GATES OF SEC. ??.

ID	Reaction	Propensity (rate law)
0.8a	$\emptyset \rightarrow H$	$k_H H_{\max}$ (zeroth order)
0.8b	$H \rightarrow \emptyset$	$k_H N_H$
0.9a	$\emptyset \rightarrow X$	$k_0 c(H) b(Rf)$ [molec s ⁻¹ cell ⁻¹]
0.9b	$2X \rightarrow 3X$	$\frac{k_1}{2} N_X (N_X - 1)$
0.9c	$3X \rightarrow 2X$	$\frac{k_6}{6} N_X (N_X - 1) (N_X - 2)$
0.9d	$X \rightarrow \emptyset$	$k_2 N_X$
0.9y1	$2X \rightarrow 2X + Y$	$\frac{k_3}{2} N_X (N_X - 1)$
0.9y2	$X + Y \rightarrow Y$	$k_5 N_X N_Y$
0.9y3	$Y \rightarrow \emptyset$	$k_4 N_Y$
0.9r1	$2X \rightarrow 2X + Rf$	$\frac{k_{Rp}}{2} N_X (N_X - 1)$
0.9r2	$Rf \rightarrow \emptyset$	$k_{Rd} N_{Rf}$
0.9e	$2X + ERK2 \rightarrow 2X + ERK2^*$	$k_{XE} [X]^2 [ERK2]$ (falling factorial in N_X)
0.9e'	$ERK2^* \rightarrow ERK2$	$k_{Ed} [ERK2^*]$
0.9f	$X + ACA \rightarrow X + ACA^*$	$k_{XA} [X] [ACA] \frac{[ERK2^*]^4}{K_E^4 + [ERK2^*]^4}$

TABLE III

MODULE 0 CONSTANTS. BASIS: L LITERATURE VALUE, D DERIVED FROM A LITERATURE ANCHOR, C CALIBRATED ON THE MODEL AGAINST A STATED TARGET, E ESTIMATE OR DESIGN CHOICE (SEC. ??).

Symbol	Value	Unit	Basis	Meaning, justification, source
$H_{\max}$	100	molec/voxel	E	Steady-state level of the abstract <i>Hunger</i> variable; held at full developmental competence.
k_H	2.8×10^{-4}	s ⁻¹	E	Relaxation of Hunger ($\tau \approx 1$ h); the production rate is $H_{\max} k_H$. A slow developmental clock, not a measured rate.
$H_{1/2}, n_c$	60, 4	molec/voxel	E	Half-competence level and steepness of the competence gate; $c(H_{\max}) = 0.885$.
k_0	3.5	molec s ⁻¹ cell ⁻¹	C	Ignition (seed) rate at full competence. Set on the stochastic engine so that the mean inter-pulse interval of an isolated cell is 726 s; target ≈ 12 min [?].
k_1	0.08	molec ⁻¹ s ⁻¹	C	Autocatalysis ($2X \rightarrow 3X$); creates the excitation threshold (≈ 23 molecules).
k_6	5×10^{-4}	molec ⁻² s ⁻¹	C	Cubic self-limitation; bounds the spike at ≈ 456 molecules per cell.
k_2	1	s ⁻¹	C	Basal removal of X (relaxation time 1 s).
k_3	6.25×10^{-5}	molec ⁻¹ s ⁻¹	C	Spike-triggered production of the recovery variable Y.
k_5	0.05	molec ⁻¹ s ⁻¹	C	Y-mediated spike termination.
k_4	0.01	s ⁻¹	C	Decay of Y ($\tau = 100$ s); sets the recovery of the element.
k_{Rp}, k_{Rd}	$5 \times 10^{-3}, 5 \times 10^{-3}$	molec ⁻¹ s ⁻¹ , s ⁻¹	C	Charging and decay ($\tau_R = 200$ s) of the refractory clock.
K_R, n_R	450, 4	molec/cell	C	Half-block level and steepness of the refractory gate (Sec. ??).
k_{XE}	1823	$\mu M^{-2} s^{-1}$	C	X ² -driven activation of ERK2 (cell units); second order in X to raise the contrast between spike and resting leak.
k_{Ed}	0.02	s ⁻¹	E	ERK2 dephosphorylation ($\tau = 50$ s); ERK2 activation is transient [?].
k_{XA}	65	$\mu M^{-1} s^{-1}$	C	X-driven ACA activation, gated on ERK2*; sets the ≈ 10 s rise of ACA*.
K_E, n_E	192, 4	molec/voxel	C	ERK2* level (12.5 % of the pool) at half-open ACA gate; enforces the ordering ERK2→ACA.
N_{vox}	13		E	Voxels per cell (plus-shaped footprint on the $2 \mu m$ lattice); $M_c = 13 \times 4817.6 = 62\,629$ molecules μM^{-1} per cell.
$[ERK2]_{\text{tot}}$	1540	molec/voxel	E	Pool size on the scale of the other kinases ($0.32 \mu M$).
$[ACA]_{\text{tot}}$	865	molec/voxel	C	Chosen so that one pulse exports $\approx 10^7$ cAMP molecules per cell at $k_9 + k_{11} = 20 \text{ s}^{-1}$ (Module 9).
$D_X, D_Y, D_{Rf}, D_{ERK2}$	10	$\mu m^2/s$	E	Generic cytosolic value; the cell is mixed in ≈ 2.5 s, which matters because the gates are evaluated per voxel.

b) Reactions.:

0.8a/b An abstract *Hunger* variable H represents the developmental state of a starved cell; it is not a molecule. It scales the competence to fire (gate $c(H)$), the

cAMP synthesis rate (9.1) and the induction of the extracellular phosphodiesterase system (Module 11). In the simulations H starts at its steady state $H_{\max}$ and fluctuates around it, so it acts as an approxi-

mately constant developmental level.

- 0.9a Spontaneous ignition: a rare, competence-gated seed that injects X, the activator, at $k_0 = 3.5$ molecules s^{-1} per cell. It is switched off by the refractory clock, which makes the block hard: with the seed shut, X decays through 0.9d and the autocatalysis 0.9b, which needs $N_X \geq 2$, has nothing to amplify.
- 0.9b–d The autocatalytic step 0.9b creates an excitation threshold, the cubic loss 0.9c bounds the spike and keeps the stochastic simulation finite, and the linear loss 0.9d sets the resting state. These three reactions form the chemical Schlögl model of a bistable/excitable system [?].
- 0.9y1–3 The recovery variable Y is produced only by spikes (quadratic in X, so resting fluctuations do not raise it), terminates the spike by removing X (0.9y2) and decays slowly (0.9y3). X and Y form the chemical counterpart of the FitzHugh–Nagumo excitable system with activator and delayed inhibitor [?], [?]. With the constants of Table ??, the cell-level nullcline gives a resting level ≈ 4 , an excitation threshold ≈ 23 and a spike maximum ≈ 456 molecules per cell.
- 0.9r1/2A refractory clock Rf is charged by spikes of X (quadratic, as for Y) and decays with $\tau_R = 200$ s. Y cannot serve as the clock because it also terminates the spike, and its time constant would then be tied to the pulse shape. Charging the clock from X makes the block *source independent*: a cell that has fired, spontaneously or after a relayed signal (Module 8), cannot be fired again by either route until the clock has decayed. The model requires a refractory period of several minutes, commensurate with the ≈ 7 min oscillation period [?] and the ≈ 4 min minimum interval between pulses of single cells [?].
- 0.9e X activates ERK2 in a second-order step. ERK2* is deliberately long-lived, so it integrates X; in the calibration a first-order drive left ERK2* at 5 % of the pool at rest, which kept the RegA brake 87 % off before any spike. The second-order dependence squares the spike-to-leak contrast and can be read as a lumped ultrasensitive step; it has no direct experimental counterpart.
- 0.9e' Basal ERK2 dephosphorylation; ERK2 activation by cAMP is rapid and transient [?].
- 0.9f X activates the adenylyl cyclase ACA once ERK2* has risen. ERK2 is required for receptor-mediated activation of adenylyl cyclase [?], and it disarms RegA (Module 10). The ERK2* gate makes ACA follow the release of the RegA brake by a few seconds, so the cAMP made by ACA* is not destroyed immediately.

c) *Constants.*: The parameters of 0.9a–d and 0.9y are calibrated, not measured: the excitable core has three targets (excitation threshold, spike height and recovery time), and the values in Table ?? reproduce them. The ignition rate k_0 is the

only parameter that sets the firing rate. It was calibrated on the stochastic engine, because neither a well-mixed reduction (about $2\times$ too slow: ignition nucleates in one voxel before diffusion levels it out) nor a 13-voxel deterministic reduction predicts it. With the refractory clock active, $k_0 = 3.5$ molecules s^{-1} gives a mean interval of 726 s (95 % interval 706–747 s, 8 seeds), close to the 12 min of [?]; the previous value 2.52 gave 930 s because the block adds about 390 s of dead time. *Limitation*: because the block is deterministic and makes up a large part of the interval, the coefficient of variation of the isolated pacemaker is 0.16, below the value $1/\sqrt{6.6} \approx 0.39$ implied by the gamma distribution measured in [?]. In populations the period converges onto the block; for 1000 cells it is 467 ± 16 s (7.8 min), of the order of the 7 min period of [?]. The clock parameters k_{Rp} , k_{Rd} , K_R were set on the measured recovery curve of the relay (no relayed pulse within 300 s of the previous pulse, one in three at 360 s, all by 420 s; three seeds), not from the analytical estimate, because the block lifts when the gated relay drive crosses the excitation threshold, which depends on the relay gain.

The Hill gates $c(H)$, $b(Rf)$ and the ERK2* gate are justified in Table ?. The choice $n = 4$ for the three is not a cooperativity claim; it makes the transition sharp enough that the gate is either nearly open or nearly shut at the values the input actually takes.

E. Module 1: cAMP Receptor Binding and Desensitisation

a) *Role.*: cAMP is sensed by the G-protein-coupled receptor cAR1. Binding to intact aggregation-competent cells is heterogeneous: about 4 % of the sites dissociate slowly ($K_d = 12.5$ nM), about 40 % are fast, high-affinity sites ($K_d = 60$ nM) and about 60 % are fast, low-affinity sites ($K_d = 450$ nM); under cAMP the high-affinity sites convert to low-affinity ones with first-order kinetics ($t_{1/2} \approx 9$ s) [?]. Occupancy induces phosphorylation of serine clusters in the C-terminal domain [?], [?], which is correlated with the loss of ligand binding and reduces the affinity of the receptor rather than removing it from the surface [?], [?]. The model resolves the two fast classes, and represents the low-affinity class as the phosphorylated receptor; the slow class (4 % of sites) is omitted. Each receptor form has the binding topology of the reduced-order receptor model of Biswas et al. [?] (Table 1, rows 27/28), $R + cAMP \rightleftharpoons RC$, to which the model adds a phosphorylated pair; the full three-class model of [?] is available as an option (Table ??) and is not used here.

b) Reactions.

- 1.1/1.2 Reversible cAMP binding to the high-affinity receptor R, with $K_d = 60$ nM. This is the fastest step of the cascade ($\tau = 1.1$ s at $c = K_d$) and is close to quasi-equilibrium with everything downstream.
- 1.3/1.4 Binding to the phosphorylated receptor R_p . Affinity is lost through the on-rate ($K_{d,p} = 480$ nM), so R_p still binds and releases cAMP, i.e. it is slower to engage, not locked out. A low-affinity receptor is still on the steep part of its binding curve at concentrations where the high-affinity receptor is saturated; R_p therefore extends the dynamic range of the stage.

TABLE IV

MODULE 1 REACTIONS (DEFAULT TWO-STATE RECEPTOR). R IS THE UNPHOSPHORYLATED CAR1 RECEPTOR, R_p ITS PHOSPHORYLATED FORM, RC AND R_pC THE CAMP-BOUND STATES; A IS THE CYTOSOLIC ADAPTER OF REACTIONS 1.8.

ID	Reaction	Propensity (rate law)
1.1	$R + \text{cAMP}_e \rightarrow RC$	$k_{\text{on}}[R][\text{cAMP}_e]$
1.2	$RC \rightarrow R + \text{cAMP}_e$	$k_{\text{off}}[RC]$
1.3	$R_p + \text{cAMP}_e \rightarrow R_pC$	$k_{\text{on},p}[R_p][\text{cAMP}_e]$
1.4	$R_pC \rightarrow R_p + \text{cAMP}_e$	$k_{\text{off},p}[R_pC]$
1.5	$RC \rightarrow R_pC$	$k_{\text{ph}}[RC]$
1.5b	$R \rightarrow R_p$	$k_{\text{ph}0}[R]$
1.6	$R_pC \rightarrow RC$	$k_{\text{dp}}[R_pC]$
1.7	$R_p \rightarrow R$	$k_{\text{b}}[R_p]$
1.8a	$RC + A \rightarrow RC + A^*$	$k_A[RC][A]$
1.8b	$A^* \rightarrow A$	$k_{A,\text{off}}[A^*]$

TABLE V

MODULE 1 CONSTANTS (DEFAULT RECEPTOR MODEL).

Symbol	Value	Unit	Basis	Meaning, justification, source
R_{tot}	3077	molec/voxel	E	40001 receptors per cell ($0.64 \mu\text{M}$). Biswas et al. use $7 \times 10^4 \pm 5 \times 10^3$ [?]; a lower aggregation-stage value was chosen here.
k_{on}	7.5	$\mu\text{M}^{-1}\text{s}^{-1}$	L	Fast high-affinity class H of Ref. [?], Table 1, taken from [?].
k_{off}	0.45	s^{-1}	L	Gives $K_d = k_{\text{off}}/k_{\text{on}} = 60 \text{ nM}$, the fast high-affinity class ($\approx 40\%$ of sites) of [?].
ε_p	0.125		C	Ratio of on-rates of R_p and R. Chosen so that $K_{d,p} = 480 \text{ nM}$, close to the 450 nM low-affinity class of [?]; the loss of affinity on phosphorylation is documented in [?], [?].
$k_{\text{on},p}$	0.938	$\mu\text{M}^{-1}\text{s}^{-1}$	D	$\varepsilon_p k_{\text{on}}$. The affinity loss is carried by the on-rate, so R_p still binds and releases cAMP.
$k_{\text{off},p}$	0.45	s^{-1}	L	Equal to k_{off} ; the fast dissociation half-life of $\approx 1 \text{ s}$ is common to both fast classes [?].
k_{ph}	0.02	s^{-1}	L	Agonist-induced phosphorylation ($t_{1/2} = 35 \text{ s}$) against $t_{1/2} \approx 45 \text{ s}$ at saturating cAMP in [?].
$k_{\text{ph}0}$	5×10^{-4}	s^{-1}	E	Ligand-independent phosphorylation ($40\times$ slower); a basal component exists [?], [?].
k_{dp}	5×10^{-3}	s^{-1}	C	Dephosphorylation of the bound receptor ($t_{1/2} = 139 \text{ s}$). Sets the phosphorylated fraction $k_{\text{ph}}/(k_{\text{ph}} + k_{\text{dp}}) = 80\%$ under sustained saturating cAMP, matching the $>80\%$ loss of high-affinity binding [?], [?].
k_{b}	5.78×10^{-3}	s^{-1}	L	$t_{1/2} = 2 \text{ min}$, the receptor dephosphorylation half-time after removal of the stimulus [?].
$k_A, k_{A,\text{off}}$	0.07, 0.02	$\mu\text{M}^{-1}\text{s}^{-1}, \text{s}^{-1}$	E	Adapter charged by RC ($\tau = 50 \text{ s}$); dormant in the default network (see text).
$[A]_{\text{tot}}$	154	molec/voxel	E	Adapter pool.
D_R	0.024	$\mu\text{m}^2/\text{s}$	L	Common lateral diffusion constant of transmembrane proteins in <i>Dictyostelium</i> , $0.024 \pm 0.004 \mu\text{m}^2/\text{s}$ ($n = 27$) [?]; Biswas et al. use 0.027 [?], [?].
D_A	10	$\mu\text{m}^2/\text{s}$	E	Generic cytosolic value.

TABLE VI

OPTIONAL FULL MULTI-AFFINITY RECEPTOR MODULE OF BISWAS ET AL. [?] (TABLE 1, ROWS 1–26), SELECTABLE IN THE CODE BUT *not* USED IN THE RESULTS OF THIS PAPER. CLASSES H, L, S: HIGH, LOW AND SLOW AFFINITY; P IS THE FREE-PHOSPHATE POOL; P MARKS PHOSPHORYLATED STATES. EACH ROW LISTS THE FORWARD AND REVERSE CONSTANTS.

ID	Reaction	Propensity (rate law)
1/2	$H + \text{cAMP} \rightleftharpoons H:C$	$k_H = 7.5, k_{-H} = 0.45$
3/4	$L + \text{cAMP} \rightleftharpoons L:C$	$k_L = 2.2, k_{-L} = 1$
5/6	$S + \text{cAMP} \rightleftharpoons S:C$	$k_S = 4, k_{-S} = 0.05$
7/8	$H \rightleftharpoons L$	$k_{HL} = 0.08, k_{-HL} = 0.0536$
9/10	$H:C \rightleftharpoons L:C$	$k_{HLC} = 0.08, k_{-HLC} = 7.1 \times 10^{-3}$
11/12	$H:C + P \rightleftharpoons P H:C$	$k_{HCP} = 5.3 \times 10^{-4}, k_{-HCP} = 8 \times 10^{-4}$
13/14	$P H + \text{cAMP} \rightleftharpoons P H:C$	$k_{PHC} = 0.04, k_{-PHC} = 2.42 \times 10^{-3}$
15/16	$H + P \rightleftharpoons P H$	$k_{PH} = 5.3 \times 10^{-4}, k_{-PH} = 8 \times 10^{-4}$
17/18	$L:C + P \rightleftharpoons P L:C$	$k_{LCP} = 5.3 \times 10^{-4}, k_{-LCP} = 8 \times 10^{-4}$
19/20	$P L + \text{cAMP} \rightleftharpoons P L:C$	$k_{PLC} = 5, k_{-PLC} = 1.5$
21/22	$L + P \rightleftharpoons P L$	$k_{PL} = 5.3 \times 10^{-4}, k_{-PL} = 8 \times 10^{-4}$
23/24	$P H \rightleftharpoons P L$	$k_{PHL} = 4.3 \times 10^{-4}, k_{-PHL} = 2.9 \times 10^{-4}$
25/26	$P H:C \rightleftharpoons P L:C$	$k_{PHLC} = 4.3 \times 10^{-4}, k_{-PHLC} = 3.8 \times 10^{-5}$

1.5/1.5b Phosphorylation, agonist-induced (1.5) and basal (1.5b, $40\times$ slower). Agonist-induced phosphorylation is measured with $t_{1/2} \approx 45 \text{ s}$ at saturating cAMP

and a ligand-independent basal component exists [?]; the model uses $t_{1/2} = 35 \text{ s}$.

1.6/1.7 Dephosphorylation of the bound (1.6) and free (1.7)

receptor. The free receptor is dephosphorylated with $t_{1/2} = 2$ min after removal of the stimulus [?], so 1.7 sets the time to resensitise after a pulse. Under sustained saturating cAMP, 80 % of receptors are phosphorylated at steady state ($k_{ph}/(k_{ph} + k_{dp})$), matching the more than 80 % loss of high-affinity sites during prolonged stimulation [?], [?].

1.8a/b A cytosolic adapter A is charged by occupied receptors and relaxes with $\tau = 50$ s. It was introduced as a fold-change-detecting gate for a direct receptor-to-cyclase edge (reaction 8.11), motivated by the fold-change response of the cAMP relay [?]. That edge is switched off ($k = 0$), so the adapter has *no downstream effect* in the default network; it is retained as an observable.

c) *Constants.*: k_{on} and k_{off} are the high-affinity constants of Table 1 of [?], which in turn come from the analysis of [?], and give $K_d = 60$ nM. The on-rate ratio $\varepsilon_p = 1/8$ makes $K_{d,p} = 480$ nM, close to the 450 nM low-affinity class; the phosphorylation-induced affinity change was reported as 3–5-fold for one low-affinity class in the literature used by [?], so the 8-fold ratio is at the upper end and equals the ratio of the two fast classes. The receptor number, 40 001 per cell, is lower than the 7×10^4 used in [?]; it is a design choice for the aggregation stage and not a re-measurement. The lateral diffusion constant $0.024 \mu\text{m}^2/\text{s}$ is the mean over 27 transmembrane proteins measured in *Dictyostelium* [?], close to the value 0.027 used by Biswas et al. [?], [?]. A much faster phospho-cycle ($k_{ph} = 1 \text{ s}^{-1}$, about 50 times faster than the measured half-time of 45 s) was tested as a way to de-saturate the receptor stage and rejected, because it made the unphosphorylated receptor resupply-limited and hence no longer a sensor; the present values are the literature-anchored ones.

d) *Limitations.*: Ligand-induced phosphorylation is not necessary for adaptation of the downstream responses or for chemotaxis; phosphorylation-deficient receptors still adapt [?]. The phospho-cycle here therefore models only the loss of binding affinity, not the termination of signalling. Identifying the low-affinity class with the phosphorylated receptor is a simplification, since the affinity conversion ($t_{1/2} \approx 9$ s [?]) is faster than phosphorylation ($t_{1/2} \approx 45$ s [?]). Receptor internalisation is not modelled, consistent with the observation that cAR1 remains at the surface [?].

F. Module 2: G-Protein Cycle

a) *Role.*: cAR1 couples to the heterotrimer $G\alpha_2\beta\gamma$. FRET between the α and β subunits shows that the heterotrimer dissociates and reassociates rapidly on addition and removal of cAMP, that during continuous stimulation the activation reaches a dose-dependent steady state and does *not* decline while the physiological responses subside, and that occupied receptors catalyse the G-protein cycle whether or not they are phosphorylated [?]. Adaptation therefore occurs downstream of the G protein, and free $G\beta\gamma$ is the persistent excitatory signal that Module 3 differentiates. The initial Ras response to uniform cAMP requires $G\beta$ but not $G\alpha_2$ [?], and

symmetry breaking and amplification occur between the G protein and Ras [?]. The module follows the G-protein module of Biswas et al. [?].

b) Reactions.:

- 2.0 Basal, receptor-independent nucleotide exchange. Its flux is negligible (about $0.03 \text{ molecules s}^{-1}$ per cell) but it defines the resting $G\beta\gamma$ floor, so that the receiver has a defined baseline rather than a hard zero.
- 2.1 Receptor-catalysed exchange: an occupied receptor dissociates the heterotrimer into $G\alpha_2$ -GTP and $G\beta\gamma$ and is not consumed.
- 2.2 The same reaction for the phosphorylated occupied receptor. Both forms catalyse the cycle with equal efficiency ($\varepsilon_E = 1$), as observed [?], so the phosphorylated receptor carries the drive at high cAMP.
- 2.3 Intrinsic GTP hydrolysis converts $G\alpha_2$ -GTP into $G\alpha_2$ -GDP. An RGS-accelerated hydrolysis (2.4) is defined in the code but switched off.
- 2.5 Recapture of $G\beta\gamma$ by $G\alpha_2$ -GDP re-forms the heterotrimer and terminates the signal.

c) *Constants.*: k_E is derived from the dissociation half-time at saturating cAMP: when all receptors are occupied the heterotrimer is consumed with the pseudo-first-order rate $k_E[R]_{\text{tot}}$, so $k_E = \ln 2/(t_{1/2,\text{diss}}[R]_{\text{tot}})$ with $t_{1/2,\text{diss}} = 3.2$ s, the value Biswas et al. reproduce from the FRET data [?], [?]. The basal constant k_{E0} is the value of Table 2 of [?] for a ratio of 3.

Why the gain of the stage was changed. The conservation law $[G\beta\gamma] = [G\alpha_2^{\text{GTP}}] + [G\alpha_2^{\text{GDP}}]$ holds exactly, because every dissociation creates one $G\alpha_2$ and one $G\beta\gamma$ and every recapture removes one of each. In the steady state, with P the total exchange flux, $[G\alpha_2^{\text{GTP}}] = P/k_{\text{hyd}}$ and $k_{\text{re}}[G\alpha_2^{\text{GDP}}][G\beta\gamma] = P$. If recapture is slow compared with hydrolysis, $[G\beta\gamma] \approx \sqrt{P/k_{\text{re}}}$ and the stage compresses the input contrast (exponent $\frac{1}{2}$); if recapture is fast, $[G\beta\gamma] \approx [G\alpha_2^{\text{GTP}}] = P/k_{\text{hyd}}$ is linear in the drive until the heterotrimer is depleted. The exponent is therefore a property of the regime and not of the conservation law. Raising k_{re} by a factor of 3500 puts the stage in the linear regime, and raising k_{hyd} by a factor of 7 moves the saturation point to higher cAMP. On the stochastic engine, with replicate cells, the contrast transfer $d \ln G\beta\gamma / d \ln c$ at 60 nM cAMP increased from 0.22 to 0.77 and the signal-to-noise ratio doubled; at 5 nM the transfer increased from 0.85 to 1.65 with no loss of signal-to-noise ratio.

Departures from the literature anchors. The chosen values move the two half-time anchors: the dissociation half-time becomes about 2 s (from 3.2 s) and the reassociation half-time about 7.5 s, four times shorter than the 30 s used by [?]. The cost of the stochastic simulation at the working point rises about threefold, because the activated fraction is increased on purpose. The G-protein pool is 60 times the receptor number, 20 times the ratio 3 of [?]. This pool size raises the $G\beta\gamma$ copy number at a fixed activated fraction and was accompanied by paired changes in Module 3 (see there) that leave its dynamics unchanged; it is a deliberate trade of realism for signal-to-noise

TABLE VII
MODULE 2 REACTIONS: THE G-PROTEIN CYCLE OF $G\alpha_2\beta\gamma$. $G\alpha_2^{\text{GTP}}$ AND $G\alpha_2^{\text{GDP}}$ ARE THE GTP- AND GDP-BOUND FREE α SUBUNITS; $G\beta\gamma$ IS THE FREE DIMER.

ID	Reaction	Propensity (rate law)
2.0	$G\alpha\beta\gamma \rightarrow G\alpha_2^{\text{GTP}} + G\beta\gamma$	$k_{E0}[G\alpha\beta\gamma]$
2.1	$G\alpha\beta\gamma + \text{RC} \rightarrow G\alpha_2^{\text{GTP}} + G\beta\gamma + \text{RC}$	$k_E[G\alpha\beta\gamma][\text{RC}]$
2.2	$G\alpha\beta\gamma + \text{R}_p\text{C} \rightarrow G\alpha_2^{\text{GTP}} + G\beta\gamma + \text{R}_p\text{C}$	$\varepsilon_E k_E[G\alpha\beta\gamma][\text{R}_p\text{C}]$
2.3	$G\alpha_2^{\text{GTP}} \rightarrow G\alpha_2^{\text{GDP}}$	$k_{\text{hyd}}[G\alpha_2^{\text{GTP}}]$
2.5	$G\alpha_2^{\text{GDP}} + G\beta\gamma \rightarrow G\alpha\beta\gamma$	$k_{\text{re}}[G\alpha_2^{\text{GDP}}][G\beta\gamma]$

TABLE VIII
MODULE 2 CONSTANTS.

Symbol	Value	Unit	Basis	Meaning, justification, source
$G:R$	60		E	G-protein to receptor ratio; Biswas et al. use 3 [?]. Chosen to raise the $G\beta\gamma$ count (shot noise) at constant activated fraction; an explicit trade of abundance realism for signal-to-noise, not a claim about abundance.
G_{tot}	184 620	molec/voxel	D	$R_{\text{tot}} \times G:R$ (38 μM).
$t_{1/2,\text{diss}}$	3.2	s	L	Dissociation half-time at saturating cAMP; the value reproduced by Biswas et al. (3.0–3.3 s, Table 2) from FRET data [?], [?].
k_E	0.339	$\mu\text{M}^{-1}\text{s}^{-1}$	D	$k_E = \ln 2 / (t_{1/2,\text{diss}} [R]_{\text{tot}})$: pseudo-first-order dissociation when all receptors are occupied.
ε_E	1		L	Phosphorylated and unphosphorylated occupied receptors catalyse the G-protein cycle equally [?]; Biswas et al. likewise weight all occupied classes equally [?].
k_{E0}	2.18×10^{-7}	s^{-1}	L	Basal, receptor-independent exchange (Table 2 of [?], value for $G:R = 3$, not rescaled); sets the resting $G\beta\gamma$ floor.
k_{hyd}	0.162	s^{-1}	C	$7 \ln 2 / 30$: intrinsic GTP hydrolysis of $G\alpha_2$, raised $7\times$ above the value matching the 30 s recovery anchor to de-saturate the stage (see text).
k_{re}	6.09	$\mu\text{M}^{-1}\text{s}^{-1}$	C	$3500 / (t_{1/2,\text{re}} [G]_{\text{tot}} / 2)$ with $t_{1/2,\text{re}} = 30$ s; the factor 3500 keeps recapture pseudo-first order (see text). The realised reassociation half-time is 7.5 s, not 30 s.
$D_{G\alpha\beta\gamma}, D_{G\alpha_2}, D_{G\beta\gamma}$	0.2	$\mu\text{m}^2/\text{s}$	E	The value $0.2 \mu\text{m}^2/\text{s}$ is that of Table 3 of [?]; equal mobility of the three forms is a design choice (G proteins also exchange between cytosol and membrane [?]).

and is not a claim about G-protein abundance. The three G-protein forms have the same diffusion constant, which is a design choice.

G. Module 3: Ras Activation, Adaptation and Amplification

a) Role.: The earliest polarised response downstream of the G protein is the activation of Ras. RasG and RasC are activated rapidly and transiently by cAMP in aggregation-competent cells [?], and RasG is the upstream activator of PI3K [?], [?]. The activation of RasG shows near-perfect adaptation over a wide range of concentrations; among the two simple topologies that can produce perfect adaptation, only an incoherent feed-forward loop fitted the measured responses, in which the receptor drives an excitatory and an inhibitory node proportionally, possibly with a RasGAP as the global inhibitor [?]. An incoherent feed-forward loop of this type has the structure of a local-excitation–global-inhibition (LEGI) gradient sensor, in which a local activator and a global inactivator are coordinately controlled by the G protein [?], [?] and can provide fold-change detection [?]. The RasGAP NF1 (DdNF1) is a major regulator of Ras activity and its loss makes Ras activity spatially and temporally unregulated [?]. Ras activation is suppressed when the cAMP concentration falls (“rectification”), which arises naturally in an incoherent feed-forward loop with zero-order ultrasensitivity [?]. Finally, the Ras/PI3K/PIP3 network is excitable: it fires without external

cues, and a stimulus biases threshold and timing [?], [?], [?], [?]. Module 3 therefore combines (i) a fast $G\beta\gamma$ -driven excitator and a slower $G\beta\gamma$ -driven global inhibitor (adaptation and spatial comparison), (ii) a saturable GAP (rectification), (iii) a PIP3-to-Ras positive feedback (amplification) and (iv) a delayed negative feedback (reset). An integral-feedback (antithetic) arm that was tested earlier was removed, because integral feedback gives oscillations and stimulus-dependent kinetics that are inconsistent with the measured Ras response [?].

b) Reactions.:

- 3.1/3.2 The excitator GEF_R is activated by free $G\beta\gamma$ and inactivated with $\tau_E = 2$ s. RasGEFR is the exchange factor of RasG [?], and the initial Ras response depends on $G\beta$ [?]. The short relaxation time makes the RasG peak follow the input within seconds.
- 3.3 The active GEF exchanges GDP for GTP on membrane-bound RasG, which produces RasG-GTP.
- 3.4e/3.4b/3.5 The inhibitor GAP is activated by $G\beta\gamma$ (3.4e), i.e. by the same variable as the excitator, and decays with $\tau_I = 30$ s (3.5). The requirement that both arms read the same input is essential: if they read different upstream variables with different dose dependence, the ratio of excitation to inhibition acquires a dose dependence of its own and the stage inverts instead of adapting (this was found when the inhibitor read the receptor occupancy). A small basal activation (3.4b) represents the stimulus-independent GAP at

TABLE IX

MODULE 3 REACTIONS IN THE DEFAULT CONFIGURATION. GEF_R AND GAP ARE THE CYTOSOLIC RASG EXCHANGE FACTOR AND GTPASE-ACTIVATING PROTEIN (ASTERISK: ACTIVE FORM); $\text{RasG}^{\text{GTP/GDP}}$ ARE THE MEMBRANE-BOUND RAS STATES; B IS THE SUBSTRATE OF THE DELAYED BRAKE (3.3C-E), ACTIVATED BY PKBR1^* FROM MODULE 8.

ID	Reaction	Propensity (rate law)
3.1	$G_{\beta\gamma} + \text{GEF}_R \rightarrow \text{GEF}_R^* + G_{\beta\gamma}$	$k_{E,\text{on}}[G_{\beta\gamma}][\text{GEF}_R]$
3.2	$\text{GEF}_R^* \rightarrow \text{GEF}_R$	$k_{E,\text{off}}[\text{GEF}_R^*]$
3.3	$\text{GEF}_R^* + \text{RasG}^{\text{GDP}} \rightarrow \text{RasG}^{\text{GTP}} + \text{GEF}_R^*$	$k_{R,\text{on}}[\text{GEF}_R^*][\text{RasG}^{\text{GDP}}]$
3.4e	$G_{\beta\gamma} + \text{GAP} \rightarrow \text{GAP}^* + G_{\beta\gamma}$	$k_{I,\text{on}}[G_{\beta\gamma}][\text{GAP}]$
3.4b	$\text{GAP} \rightarrow \text{GAP}^*$	$k_{I,\text{b}}[\text{GAP}]$
3.4f	$R_pC + \text{GAP} \rightarrow \text{GAP}^* + R_pC$	$k_{I,p}[R_pC][\text{GAP}] \quad (k_{I,p} = 0)$
3.5	$\text{GAP}^* \rightarrow \text{GAP}$	$k_{I,\text{off}}[\text{GAP}^*]$
3.7z	$\text{GAP}^* + \text{RasG}^{\text{GTP}} \rightarrow \text{RasG}^{\text{GDP}} + \text{GAP}^*$	$k_{\text{cat}}[\text{GAP}^*] \frac{[\text{RasG}^{\text{GTP}}]}{K_{\text{NF1}} + [\text{RasG}^{\text{GTP}}]}$
3.3b	$\text{RasG}^{\text{GDP}} + \text{PIP3} \rightarrow \text{RasG}^{\text{GTP}} + \text{PIP3}$	$k_{\text{PIP3}}[\text{PIP3}][\text{RasG}^{\text{GDP}}]$
3.3c	$\text{PKBR1}^* + B \rightarrow B^* + \text{PKBR1}^*$	$k_{B,\text{on}}[\text{PKBR1}^*][B]$
3.3d	$B^* \rightarrow B$	$k_{B,\text{off}}[B^*]$
3.3e	$B^* + \text{RasG}^{\text{GTP}} \rightarrow \text{RasG}^{\text{GDP}} + B^*$	$k_{B,\text{hyd}}[B^*][\text{RasG}^{\text{GTP}}]$

TABLE X

MODULE 3 CONSTANTS. RATE CONSTANTS AND POOL SIZES ARE PAIRED: $k_{E,\text{on}}[\text{GEF}_R]_{\text{tot}}$, $k_{R,\text{on}}[\text{GEF}_R]_{\text{tot}}$ AND $k_{I,\text{on}}[\text{GAP}]_{\text{tot}}$ WERE HELD FIXED WHEN THE POOLS WERE ENLARGED, SO THAT ONLY SHOT NOISE CHANGED.

Symbol	Value	Unit	Basis	Meaning, justification, source
$[\text{RasG}]_{\text{tot}}$	38 500	molec/voxel	E	5×10^5 per cell; large for low shot noise. The copy number is not well measured.
$[\text{GEF}_R]_{\text{tot}}$	138 600	molec/voxel	E	Excitor pool (1.8×10^6 per cell). RasGEFR activates RasG [?].
$[\text{GAP}]_{\text{tot}}$	46 200	molec/voxel	E	Inhibitor pool (6×10^5 per cell); a RasGAP (NF1) is the global inhibitor [?].
$k_{E,\text{on}}$	3.8×10^{-3}	$\mu\text{M}^{-1}\text{s}^{-1}$	C	$G_{\beta\gamma}$ -driven GEF activation; the initial Ras response requires G_{β} [?].
$k_{E,\text{off}}$	0.5	s^{-1}	C	GEF inactivation, $\tau_E = 2$ s: the excitor is fast, so the Ras peak occurs within seconds [?], [?].
$k_{R,\text{on}}$	0.333	$\mu\text{M}^{-1}\text{s}^{-1}$	C	GEF-catalysed GDP→GTP exchange on RasG.
$k_{I,\text{on}}$	2.53×10^{-4}	$\mu\text{M}^{-1}\text{s}^{-1}$	C	$G_{\beta\gamma}$ -driven GAP activation. It reads the <i>same</i> input as 3.1, as an incoherent feed-forward loop requires [?], [?].
$k_{I,\text{off}}$	0.033	s^{-1}	C	GAP inactivation, $\tau_I = 30$ s: the inhibitor is slower than the excitor, which produces adaptation [?].
$k_{I,\text{b}}$	5×10^{-4}	s^{-1}	C	Stimulus-independent GAP activity. Without it the resting Ras/PIP3 loop latches; consistent with a basal RasGAP [?], [?]. Minimum value giving the low rest state.
$k_{I,p}$	0	$\mu\text{M}^{-1}\text{s}^{-1}$	E	GAP activation by R_pC is defined but switched off: it reads a different upstream variable than 3.1 and inverts the response above ≈ 10 nM.
K_{NF1}	0.2	μM	L	Michaelis constant of the NF1 GAP domain, $0.3 \mu\text{M}$ [?], [?]; the value is rounded to 0.2. Mammalian catalytic domain; no <i>Dictyostelium</i> kinetics exist.
k_{cat}	15.8	s^{-1}	D	Derived from the resting distance to the zero-order switch, $\theta_{\text{rest}} = 0.5$. Only the product $k_{\text{cat}}[\text{GAP}^*]$ enters; it is $11\times$ the mammalian $k_{\text{cat}} = 1.4 \text{ s}^{-1}$, so the active-GAP pool stands for more enzyme than is measured.
k_{PIP3}	0.04	$\mu\text{M}^{-1}\text{s}^{-1}$	C	Ras/PIP3 positive feedback; the amplifier of the excitable network [?], [?], [?]. Expressed relative to the legacy PIP2 pool (Sec. ??).
$k_{B,\text{on}}$	24.1	$\mu\text{M}^{-1}\text{s}^{-1}$	D	Brake activation by PKBR1^* ; half-activation at 20 PKBR1^* molecules/voxel.
$k_{B,\text{off}}$	0.1	s^{-1}	E	Brake decay ($\tau = 10$ s); sets the refractory time of the Ras/PIP3 loop.
$k_{B,\text{hyd}}$	39.8	$\mu\text{M}^{-1}\text{s}^{-1}$	D	Chosen so that at half activation the brake removes RasG-GTP at $0.6\times$ the rate of the LEGI GAP arm at the driven state (design target).
$[B]_{\text{tot}}$	1540	molec/voxel	E	Brake pool; the mean braking action is independent of pool size, so it was raised to reduce noise.
$D_{\text{GEF}_R}, D_{\text{RasG}}$	0.1	$\mu\text{m}^2/\text{s}$	E	Local excitor and membrane-bound Ras (prenylated anchor).
D_{GAP}, D_B	20	$\mu\text{m}^2/\text{s}$	E	Global inhibitor and global brake. Cytosolic value; for B^* , a membrane-attached species in the code, it is a caveat of the present model.

rest, which is needed because the resting Ras/PIP3 loop otherwise latches on; a basal RasGAP is consistent with the enhanced basal Ras activity of GAP-deficient cells [?], [?].

3.4f GAP activation by the phosphorylated receptor. It is defined and switched off ($k_{I,p} = 0$): it reads a different upstream variable from 3.1 and inverted the

response above about 10 nM.

3.7z The active GAP hydrolyses RasG-GTP with Michaelis–Menten kinetics. Because K_{NF1} is far below the Ras pool, the enzyme operates in the zero-order regime, where a graded input is turned into a switch-like output and the response to a falling stimulus is suppressed [?]. The Michaelis constant

of the NF1 GAP domain is $0.3 \mu\text{M}$ and its k_{cat} is 1.4 s^{-1} (mammalian catalytic domain) [?], [?]. The plain mass-action form of this step is available and is not used.

- 3.3b PIP3 promotes the exchange on RasG. Together with 4.1 (RasG-GTP recruits PI3K) and 4.3 (PI3K makes PIP3) this closes the Ras→PI3K→PIP3→Ras positive-feedback loop, which is the amplifier of the excitable network [?], [?]. The LEGI comparison of 3.1–3.7 provides the spatial signal but no amplification; the amplification sits between the G protein and Ras [?].

- 3.3c/d/a delayed negative feedback that terminates the patch and prevents the positive feedback from latching. In cells lacking PKB and PKBR1, RasC activation is increased and fails to adapt by 40 s [?], and the negative regulation by the PKBs extends to other Ras proteins and to PI3K, which makes them candidates for the delayed negative feedback of the excitable network [?]. The brake substrate B is a stand-in for the Sca1-like PKB substrate: it is activated by the membrane kinase PKBR1* of Module 8 (3.3c), decays with $\tau = 10 \text{ s}$ (3.3d) and accelerates the removal of RasG-GTP (3.3e). PKB phosphorylates the scaffold Sca1 and thereby controls RasC [?]; the mechanism for the RasG reporter is not identified [?], so applying B to RasG is a lumped assumption.

c) *Constants.*: The rate constants of the LEGI arms (3.1–3.5) are constrained by dynamical targets rather than by single measurements: an excitor time constant of 2 s, an inhibitor time constant of 30 s and near-perfect adaptation of the plateau to a sustained step. The pool sizes were increased to reduce shot noise, and the rate constants were rescaled so that only the products $k_{E,\text{on}}[\text{GEFR}]_{\text{tot}}$, $k_{R,\text{on}}[\text{GEFR}]_{\text{tot}}$ and $k_{I,\text{on}}[\text{GAP}]_{\text{tot}}$ enter the dynamics; the pool enlargement in Module 2 was paired with rescaling of $k_{E,\text{on}}$ and $k_{I,\text{on}}$ so that the activated fractions and time constants do not change. Unpaired, the enlargement makes the stage worse, since the activated fraction of the excitor changes with the $G\beta\gamma$ level. K_{NF1} is taken from the mammalian catalytic domain because no *Dictyostelium* kinetics exist; the k_{cat} of the model is derived from the distance to the zero-order switch and is 11 times the mammalian value, so the active-GAP pool stands for more enzyme than is measured. The feedback gain k_{PIP3} cannot be raised as a means to improve gradient contrast: because the feedback acts on the level, it lifts the resting baseline as much as the driven response, and a contrast is a ratio; above a critical gain the loop is bistable and stops resetting. The brake constants are design targets, not measurements: the activation is derived from a half-activation level of 20 PKBR1* molecules per voxel and the removal rate from a target strength (0.6 of the LEGI GAP arm), so that the brake is off at rest and engages within a patch. The brake and the GAP have a cytosolic diffusion constant of $20 \mu\text{m}^2/\text{s}$, which is appropriate for a cytosolic GAP but not for the membrane-associated brake B* of the code; it makes the brake global rather than local, so that it subtracts the same amount everywhere and leaves the

spatial contrast standing, and this is a caveat of the present parameterisation. The saturable GAP is discussed in Table ??.

H. Module 4: PIP3, PTEN and the PkbA Feedback

a) *Role.*: In chemotaxing cells PI3K moves to the plasma membrane at the front and PTEN shows a reciprocal distribution at the back [?], [?]. Bulk PIP3 rises transiently after a stimulus, PI3K activation increases within 5 s and then declines, and the changes are larger and longer in cells lacking PTEN [?]. PIP3 is read out by PH-domain proteins such as CRAC, whose translocation reflects activation of the G-protein pathway [?]; after uniform stimulation PHCrac-GFP translocation peaks after 6–8 s and disappears after about 20 s, followed by small patches from about 30 s [?]. Ras activates PI3K [?], [?], and PIP3 in turn suppresses the membrane binding of PTEN, so that PIP3 and PTEN are mutually inhibitory and can form a bistable pair [?]. A delayed negative feedback that reduces Ras and PI3K activity is provided by the PKBs, whose activation lags behind that of Ras and PI3K [?], [?]. Module 4 implements these elements with a conserved lipid pool, a conserved PTEN pool and a two-step PkbA arm.

b) *Reactions.*:

- 4.1/4.2 RasG-GTP recruits cytosolic PI3K to the membrane, and membrane PI3K is released again. The release rate sets how fast PI3K activity declines after the stimulus [?].
- 4.3 Membrane PI3K converts PIP2 to PIP3. The flux is proportional to the membrane PI3K count and does not depend on PIP3, so the rate constant sets the PIP3 count and not its contrast.
- 4.4/4.5 PTEN docks on membrane PIP2 and is released; this gives PTEN its steady membrane fraction.
- 4.6c PIP3 evicts PTEN from the membrane through a cooperative gate; this is the mutual inhibition PIP3 $\dashv$ PTEN [?]. The Hill-type form is used, instead of a mass-action form, because it separates a low-PIP3 state (PTEN on the membrane) from a high-PIP3 state (PTEN off the membrane).
- 4.7 Membrane PTEN removes the 3-phosphate of PIP3 [?], [?].
- 4.8 A PTEN-independent phosphatase (a 5-phosphatase such as SHIP) removes PIP3. It is the dominant sink at rest: without it PTEN, which is evicted by PIP3, would be the only route out of PIP3 and the resting cell would convert 60 % of the lipid pool to PIP3. The two-step dephosphorylation via PI(3,4)P₂ is collapsed to one step, which preserves the flux out of PIP3 but does not resolve PI(3,4)P₂, a species that is itself part of the Ras excitable network [?].
- 4.10/4. PIP3 recruits the kinase PkbA (Akt) by its PH domain, and PkbA is released from the membrane. PkbA is recruited to the membrane in response to cAMP in a PI3K-dependent way and its activation is transient [?].
- 4.12/4. Membrane PkbA is activated by TORC2, which is active at the front and acts through the hydrophobic motif [?], and inactivated with $\tau = 50 \text{ s}$.

TABLE XI

MODULE 4 REACTIONS: PI3K RECRUITMENT, PIP3/PTEN MUTUAL INHIBITION, PIP3 REMOVAL, AND THE DELAYED PKBA FEEDBACK. SUBSCRIPTS C AND M DENOTE CYTOSOLIC AND MEMBRANE-BOUND FORMS; N IS A MOLECULE COUNT PER VOXEL.

ID	Reaction	Propensity (rate law)
4.1	$\text{RasG}^{\text{GTP}} + \text{PI3K}_c \rightarrow \text{PI3K}_m + \text{RasG}^{\text{GTP}}$	$k_{3K,\text{on}}[\text{RasG}^{\text{GTP}}][\text{PI3K}_c]$
4.2	$\text{PI3K}_m \rightarrow \text{PI3K}_c$	$k_{3K,\text{off}}[\text{PI3K}_m]$
4.3	$\text{PIP2} + \text{PI3K}_m \rightarrow \text{PIP3} + \text{PI3K}_m$	$k_{\text{syn}}[\text{PIP2}][\text{PI3K}_m]$
4.4	$\text{PTEN}_c + \text{PIP2} \rightarrow \text{PTEN}_m + \text{PIP2}$	$k_{PT,\text{on}}[\text{PTEN}_c][\text{PIP2}]$
4.5	$\text{PTEN}_m \rightarrow \text{PTEN}_c$	$k_{PT,\text{off}}[\text{PTEN}_m]$
4.6c	$\text{PTEN}_m \rightarrow \text{PTEN}_c$	$k_{\text{ev}}[\text{PTEN}_m] \frac{N_{\text{PIP3}}^2}{K_{\text{ev}}^2 + N_{\text{PIP3}}^2}$
4.7	$\text{PIP3} + \text{PTEN}_m \rightarrow \text{PIP2} + \text{PTEN}_m$	$k_{PT}[\text{PIP3}][\text{PTEN}_m]$
4.8	$\text{PIP3} \rightarrow \text{PIP2}$	$k_{\text{SH}}[\text{PIP3}]$
4.10	$\text{PIP3} + \text{PKBA}_c \rightarrow \text{PKBA}_m + \text{PIP3}$	$k_{K,\text{on}}[\text{PIP3}][\text{PKBA}_c]$
4.11	$\text{PKBA}_m \rightarrow \text{PKBA}_c$	$k_{K,\text{off}}[\text{PKBA}_m]$
4.12	$\text{PKBA}_m + \text{TORC2}^* \rightarrow \text{PKBA}_m^* + \text{TORC2}^*$	$k_{K,\text{act}}[\text{PKBA}_m][\text{TORC2}^*]$
4.13	$\text{PKBA}_m^* \rightarrow \text{PKBA}_c$	$k_{K,\text{deact}}[\text{PKBA}_m^*]$
4.14	$\text{PKBA}_m^* + \text{PI3K}_m \rightarrow \text{PKBA}_m^* + \text{PI3K}_c$	$k_{K,\text{fb}}[\text{PKBA}_m^*][\text{PI3K}_m]$

TABLE XII

MODULE 4 CONSTANTS. ALL RATES EXCEPT THE PKBA ARM (4.10–4.14) CARRY THE UNIFORM TIME-SCALE FACTOR $s_4 = 20$ (SEC. ??).

Symbol	Value	Unit	Basis	Meaning, justification, source
s_4	20		C	Uniform clock multiplier on all Module 4 rates. Leaves every steady state and loop gain unchanged; it makes PIP3 formation complete within seconds of a 40 s stimulus.
$[\text{PI3K}]_{\text{tot}}$	200 000	molec/voxel	E	Pool large enough that catalyst depletion is negligible.
$[\text{PIP2}]_{\text{tot}}$	769 231	molec/voxel	E	Lipid pool (10^7 per cell), conserved: $\text{PIP2} + \text{PIP3} = \text{const.}$
$[\text{PTEN}]_{\text{tot}}$	10 000	molec/voxel	E	PTEN pool, conserved between cytosol and membrane.
$[\text{PKBA}]_{\text{tot}}$	100 000	molec/voxel	C	Pool of the PH-domain kinase PkbA; enlarged so that its feedback can act at low activated fractions.
$k_{3K,\text{on}}$	0.4	$\mu\text{M}^{-1}\text{s}^{-1}$	C	RasG-GTP recruits PI3K to the membrane [?], [?].
$k_{3K,\text{off}}$	10	s^{-1}	C	PI3K membrane release; PI3K activity is transient after a stimulus [?].
k_{syn}	0.0955	$\mu\text{M}^{-1}\text{s}^{-1}$	C	PIP3 synthesis per membrane PI3K; sets the PIP3 count (shot noise), not its contrast.
$k_{PT,\text{on}}, k_{PT,\text{off}}$	0.6, 4	$\mu\text{M}^{-1}\text{s}^{-1}, \text{s}^{-1}$	C	PTEN docking on PIP2 and release; PTEN sits on the membrane at the back of the cell [?], [?].
k_{ev}	800	s^{-1}	C	Saturated PIP3-dependent eviction rate of PTEN.
$K_{\text{ev}}, n_{\text{ev}}$	$1.31 \times 10^5, 2$	molec/voxel, –	C	Half-eviction level (17 % of the lipid pool) and steepness of the eviction gate (Sec. ??).
k_{PT}	1.39	$\mu\text{M}^{-1}\text{s}^{-1}$	C	PTEN dephosphorylation of PIP3 to PIP2 [?], [?].
k_{SH}	1.28	s^{-1}	C	PTEN-independent 5-phosphatase removal of PIP3; $k_{\text{SH}} = k_{\text{ship}}[\text{SHIP}]$ with a constant catalyst pool of 77 molecules/voxel.
$k_{K,\text{on}}$	1.32×10^{-3}	$\mu\text{M}^{-1}\text{s}^{-1}$	D	PH-domain recruitment of PkbA; solved so that the resting rate is 0.002 s^{-1} at 7298 PIP3/voxel.
$k_{K,\text{off}}$	0.05	s^{-1}	E	PkbA membrane release.
$k_{K,\text{act}}$	0.507	$\mu\text{M}^{-1}\text{s}^{-1}$	D	TORC2-dependent activation; solved so that the resting rate is $8.5 \times 10^{-3} \text{ s}^{-1}$ at 81 TORC2*/voxel.
$k_{K,\text{deact}}$	0.02	s^{-1}	E	PkbA* decay ($\tau = 50 \text{ s}$); sets the refractory duration.
$k_{K,\text{fb}}$	160	$\mu\text{M}^{-1}\text{s}^{-1}$	C	Strength of the PKB feedback on PI3K, chosen so that PIP3 returns to baseline after a pulse.
$D_{\text{PI3K}_c}, D_{\text{PTEN}_c}, D_{\text{PKBA}_c}$	10	$\mu\text{m}^2/\text{s}$	E	Cytosolic value; the PTEN shuttle crosses the cell in $\approx 2.5 \text{ s}$.
$D_{\text{PI3K}_m}, D_{\text{PTEN}_m}, D_{\text{PIP2/3}}, D_{\text{PKBA}_m}$	0.1	$\mu\text{m}^2/\text{s}$	E	Membrane-bound species; the active PkbA is kept local so that feedback ends the patch where it forms.

The two-step activation (PIP3-dependent recruitment first, TORC2-dependent phosphorylation second) is a coincidence detector and provides the lag behind PIP3 that a delayed negative feedback needs. The phosphoinositide-dependent kinases PdkA and PdkB, which phosphorylate the activation loop [?], are

not modelled; activation-loop and hydrophobic-motif phosphorylation are lumped into one step.

4.14 Active PkbA removes PI3K from the membrane. The literature target of the feedback is the Sca1/RasGEF/PP2A complex and thus RasC [?]; because that complex is not resolved, the feedback

is applied at its Module 4 end point (less Ras-GTP implies less membrane PI3K). Sign, locality and delay follow the published mechanism, and only the intermediate is omitted.

PkbA (with a PH domain, recruited by PIP3) and PKBR1 (myristoylated, constitutively at the membrane, PI3K-independent) are two different gene products [?], [?]; Module 4 owns PkbA and Module 8 owns PKBR1, so that no reaction describes the same species twice.

c) *Constants.*: All rates of reactions 4.1–4.8 are multiplied by the clock factor $s_4 = 20$. A uniform factor on all rates in a module is a clock, not a re-parameterisation: every steady state, every activated fraction and the loop gain are unchanged, and only the time scales are divided. What limits PIP3 formation is the flux against the pool that has to be converted (about 1.7×10^3 molecules per voxel per second against a pool of about 6×10^4), and the factor was chosen so that PIP3 forms within seconds of a 40 s stimulus. The constants k_{syn} , $k_{PT,\text{on}}$ and k_{PT} are expressed relative to legacy pool sizes (divided by the ratio of the present pool to the legacy pool), so that the flux per pool does not change when the pools were enlarged. Enlarging a production-rate constant raises resting and peak PIP3 together, so it changes the PIP3 count (and hence the noise), not the contrast; it was therefore used as a noise lever and not as a contrast lever. $k_{K,\text{on}}$ and $k_{K,\text{act}}$ are solved from resting-rate anchors (0.002 and $8.5 \times 10^{-3} \text{ s}^{-1}$), so that the resting rate of the feedback arm is fixed and does not silently rescale when Modules 3 and 8 are retuned. The slow recruitment ($k_{K,\text{on}}$) delays the feedback, so that it acts after the cAMP pulse and not during it. The PIP3 gate K_{ev} was set on the steep flank of the eviction curve; a value of the half-point above 19.5 % of the lipid pool made the PIP3 domain a winner-take-all latch whose orientation was set by the initial condition and not by the gradient.

d) *Limitations.*: The mutual inhibition of Module 4 alone does not amplify a spatial signal: for the pools used its loop gain stays below one, so that the amplification of the polarity signal is provided by the Ras→PIP3 feedback 3.3b and not by PTEN. The PkbA feedback is required for a reset after a pulse: without it PIP3 keeps rising after the stimulus has ended. The gate of reaction 4.12 reads TORC2, which is activated by RasC in the literature [?]; the coupling of Module 8 to Module 4 through TORC2* follows this.

I. Module 8: RasC–TORC2–PKB–CRAC Relay

a) *Role.*: The relay of the cAMP signal, i.e. the re-synthesis of cAMP in response to cAMP, runs through a second Ras branch. RasC and RasG are both activated by cAMP, but RasC is responsible for adenylyl cyclase activation and RasG for chemotaxis; RasGEFA activates RasC but not RasG [?]. RasC is an upstream regulator of the target of rapamycin complex 2 (TORC2), which phosphorylates the myristoylated kinase PKBR1 independently of PIP3 and prolongs the time course of ACA activation [?]. Within seconds of a stimulus, TORC2 activates two PKB homologues, PkbA and PKBR1, also in cells lacking PI3K activity [?]. PKBR1 has no PH domain, is myristoylated for constitutive membrane

localisation and does not require PI3K [?]. PKB and PKBR1 phosphorylate the scaffold Sca1 of a Sca1/RasGEF/PP2A complex and thereby control RasC in a negative feedback [?], and PKA exerts a further negative feedback on Ras, Rap1 and TORC2 signalling [?]. The effector CRAC (a PH-domain protein) is required for receptor- and G-protein-mediated activation of ACA [?]; CRAC binds PI3K products [?] and is the translocation reporter of the G-protein pathway [?]. The model connects this chain to the excitable element of Module 0 by an output edge (8.14) that *injects the excitable activator X*, and not by a direct activation of ACA. A direct receptor-to-ACA edge would make the receptor arm a second pulse generator with its own amplitude and refractory state, whereas injecting X makes a relayed pulse identical to a spontaneous one and lets one refractory clock (Module 0) gate both.

b) Reactions.

- 8.1/8.2 RasGEFA is activated by $G\beta\gamma$ and inactivated with $\tau = 10 \text{ s}$. It uses the same $G\beta\gamma$ -dependent activated-fraction curve as the RasG excitor 3.1, since both are $G\beta\gamma$ -driven RasGEFs [?].
- 8.2s The PKB-phosphorylated Sca1-like brake substrate B* (Module 3) releases active RasGEFA, which is the negative feedback of [?]. Without PKB or TORC2, RasC activation is increased and fails to adapt by 40 s; the constant k_S is calibrated to return RasC near baseline within about 40 s.
- 8.3/8.4 Active RasGEFA converts RasC-GDP to RasC-GTP; RasC-GTP hydrolyses slowly.
- 8.4b–dA receptor-driven, slowly decaying RasC GAP (RC → CGAP*, decay $\tau = 200 \text{ s}$) hydrolyses RasC-GTP and holds RasC in its adapted state.
- 8.5/8.6 RasC-GTP activates TORC2 [?], which relaxes with $\tau = 2 \text{ s}$. The short lifetime is a design choice that places the peak of TORC2 targets within 5–10 s, as observed for the PKB substrate Sca1 [?].
- 8b.1/8b.2 TORC2* phosphorylates the membrane-anchored PKBR1 directly (no recruitment step), and PKBR1* is dephosphorylated rapidly, so that PKBR1 is a fast-on/fast-off pulse generator that is cleared before the slow negative feedbacks accumulate.
- 8b.4 PKA* inactivates PKBR1*, which closes a slow negative feedback from cAMP to the relay arm [?].
- 8.9, 8b.3 Both PKBs phosphorylate the membrane-bound CRAC pool, PkbA (Module 4) through 8.9 and PKBR1 through 8b.3.
- 8.9b/8.9c/8.10b CRAC is a three-state moiety: cytosolic CRAC_c, membrane-bound CRAC_m and phosphorylated CRAC_m*. Two facts must hold together and are compatible only if recruitment and activation are separate steps: CRAC is a PH-domain protein recruited by PIP3 [?], [?], and PKB signalling and the relay do not require PIP3 [?]. PIP3 therefore recruits CRAC (8.9b), a PIP3-independent association keeps the relay functional without PIP3 (8.9c), and the kinases phosphorylate CRAC on the membrane (8.9, 8b.3). Dephosphorylation (8.10) returns CRAC* to the membrane pool and not to the cytosol, because a phosphatase does not strip the

TABLE XIII

MODULE 8 REACTIONS: THE RAS $\rightarrow$ TORC2 $\rightarrow$ PKB $\rightarrow$ CRAC RELAY ARM AND ITS OUTPUT EDGE 8.14 INTO THE EXCITABLE ELEMENT OF MODULE 0. PKBR1 (MODULE 8B) AND CRAC ARE THE KINASE AND EFFECTOR; $g(\text{PIP3}) = [1 + (K_P/N_{\text{PIP3}})^8]^{-1}$ AND $b(\text{Rf})$ ARE THE GATES OF SEC. ??.

ID	Reaction	Propensity (rate law)
8.1	$G_{\beta\gamma} + \text{GEF}_A \rightarrow \text{GEF}_A^* + G_{\beta\gamma}$	$k_{A,\text{on}}[G_{\beta\gamma}][\text{GEF}_A]$
8.2	$\text{GEF}_A^* \rightarrow \text{GEF}_A$	$k_{A,\text{off}}[\text{GEF}_A^*]$
8.2s	$B^* + \text{GEF}_A^* \rightarrow \text{GEF}_A + B^*$	$k_S[B^*][\text{GEF}_A^*]$
8.3	$\text{GEF}_A^* + \text{RasC}^{\text{GDP}} \rightarrow \text{RasC}^{\text{GTP}} + \text{GEF}_A^*$	$k_{C,\text{on}}[\text{GEF}_A^*][\text{RasC}^{\text{GDP}}]$
8.4	$\text{RasC}^{\text{GTP}} \rightarrow \text{RasC}^{\text{GDP}}$	$k_{C,\text{off}}[\text{RasC}^{\text{GTP}}]$
8.4b	$\text{RC} + \text{CGAP} \rightarrow \text{RC} + \text{CGAP}^*$	$k_{CG,\text{on}}[\text{RC}][\text{CGAP}]$
8.4c	$\text{CGAP}^* \rightarrow \text{CGAP}$	$k_{CG,\text{off}}[\text{CGAP}^*]$
8.4d	$\text{CGAP}^* + \text{RasC}^{\text{GTP}} \rightarrow \text{RasC}^{\text{GDP}} + \text{CGAP}^*$	$k_{CG,\text{hyd}}[\text{CGAP}^*][\text{RasC}^{\text{GTP}}]$
8.5	$\text{RasC}^{\text{GTP}} + \text{TORC2} \rightarrow \text{TORC2}^* + \text{RasC}^{\text{GTP}}$	$k_{T,\text{on}}[\text{RasC}^{\text{GTP}}][\text{TORC2}]$
8.6	$\text{TORC2}^* \rightarrow \text{TORC2}$	$k_{T,\text{off}}[\text{TORC2}^*]$
8b.1	$\text{TORC2}^* + \text{PKBR1} \rightarrow \text{PKBR1}^* + \text{TORC2}^*$	$k_{P,\text{on}}[\text{TORC2}^*][\text{PKBR1}]$
8b.2	$\text{PKBR1}^* \rightarrow \text{PKBR1}$	$k_{P,\text{off}}[\text{PKBR1}^*]$
8b.3	$\text{PKBR1}^* + \text{CRAC}_m \rightarrow \text{CRAC}_m^* + \text{PKBR1}^*$	$k_{Cr}[\text{PKBR1}^*][\text{CRAC}_m]$
8b.4	$\text{PKA}^* + \text{PKBR1}^* \rightarrow \text{PKA}^* + \text{PKBR1}$	$k_{PKA}[\text{PKA}^*][\text{PKBR1}^*]$
8.9	$\text{PKBA}_m^* + \text{CRAC}_m \rightarrow \text{CRAC}_m^* + \text{PKBA}_m^*$	$k_{Cr}[\text{PKBA}_m^*][\text{CRAC}_m]$
8.9b	$\text{PIP3} + \text{CRAC}_c \rightarrow \text{CRAC}_m + \text{PIP3}$	$k_{Cr,P}[\text{PIP3}][\text{CRAC}_c]$
8.9c	$\text{CRAC}_c \rightarrow \text{CRAC}_m$	$k_{Cr,b}[\text{CRAC}_c]$
8.10	$\text{CRAC}_m^* \rightarrow \text{CRAC}_m$	$k_{Cr,\text{dp}}[\text{CRAC}_m^*]$
8.10b	$\text{CRAC}_m \rightarrow \text{CRAC}_c$	$k_{Cr,\text{off}}[\text{CRAC}_m]$
8.14	$\text{CRAC}_m^* \rightarrow \text{CRAC}_m^* + X$	$\frac{k_X}{N_{\text{CRAC}}} N_{\text{CRAC}^*} g(\text{PIP3}) b(\text{Rf}) \quad [\text{molec s}^{-1} \text{cell}^{-1}]$

TABLE XIV

MODULE 8 CONSTANTS (A): RAS C BRANCH, TORC2 AND PKBR1.

Symbol	Value	Unit	Basis	Meaning, justification, source
$[\text{GEF}_A]_{\text{tot}}$	1600	molec/voxel	D	Sized so that the number of active RasGEFA at the driven state (67 000 $G_{\beta\gamma}$ /voxel) equals that of the previous, saturated parameterisation.
$k_{A,\text{on}}$	7.6×10^{-4}	$\mu\text{M}^{-1}\text{s}^{-1}$	D	$k_{E,\text{on}}k_{A,\text{off}}/k_{E,\text{off}}$: RasGEFA and RasGEFR are both $G_{\beta\gamma}$ -driven GEFs [?] and share the activated-fraction curve in $G_{\beta\gamma}$.
$k_{A,\text{off}}$	0.1	s^{-1}	E	RasGEFA inactivation.
k_S	1	$\mu\text{M}^{-1}\text{s}^{-1}$	C	Release of active RasGEFA by the PKB-phosphorylated Sca1-like brake (negative feedback on RasC); calibrated so that RasC returns near baseline within ≈ 40 s [?].
$k_{C,\text{on}}$	2	$\mu\text{M}^{-1}\text{s}^{-1}$	E	RasGEFA-catalysed exchange on RasC [?].
$k_{C,\text{off}}$	0.02	s^{-1}	E	Intrinsic RasC hydrolysis (constitutive, slow).
$k_{CG,\text{on}}, k_{CG,\text{off}}, k_{CG,\text{hyd}}$	0.05, 0.005, 150	$\mu\text{M}^{-1}\text{s}^{-1}, \text{s}^{-1}, \mu\text{M}^{-1}\text{s}^{-1}$	E	Receptor-driven, slowly decaying RasC GAP that holds RasC adapted.
$k_{T,\text{on}}$	50	$\mu\text{M}^{-1}\text{s}^{-1}$	C	TORC2 activation by RasC-GTP [?].
$k_{T,\text{off}}$	0.5	s^{-1}	C	TORC2 inactivation ($\tau = 2$ s). $k_{T,\text{on}}$ and $k_{T,\text{off}}$ were scaled together, which leaves the activated fraction unchanged and shortens only the latency to the observed 5–10 s PKB-substrate peak [?], [?].
$k_{P,\text{on}}, k_{P,\text{off}}$	30, 0.5	$\mu\text{M}^{-1}\text{s}^{-1}, \text{s}^{-1}$	E	PKBR1 activation by TORC2 and its rapid dephosphorylation ($\tau = 2$ s); PKBR1 is constitutively membrane-bound and PI3K-independent [?], [?].

lipid anchor; releasing it to the cytosol would cap the activated fraction at $k_{\text{rec}}/(k_{\text{rec}} + k_{\text{off}})$, whatever the kinase drive. Release from the membrane (8.10b) is a separate step.

- 8.14 CRAC* injects X into the excitable element at up to $k_X = 200$ molecules s^{-1} per cell, gated by PIP3 and by the refractory clock. The PIP3 gate requires that the cell has already resolved the direction of the signal before it re-amplifies it: PIP3 is the first species of the cascade that is both stimulus-driven

and polarised.

Three further edges exist in the code and are excluded from the default network: the activation of PkbA by TORC2 without a recruitment step (8.7/8.8) is PKBR1's mechanism applied to PkbA's species and would double count the kinase that Module 8b already provides; the PKA feedback on PkbA (8.13) is excluded together with them; and the direct activation of ACA by CRAC* (8.11) is disabled ($k = 0$) for the reason stated above.

TABLE XV
MODULE 8 CONSTANTS (B): CRAC, THE RELAY OUTPUT EDGE, POOLS AND DIFFUSION CONSTANTS.

Symbol	Value	Unit	Basis	Meaning, justification, source
k_{Cr}	10	$\mu\text{M}^{-1}\text{s}^{-1}$	C	Phosphorylation of membrane-bound CRAC by either PKB. Both PKBs are TORC2 targets [?]; and RasC-TORC2-PKBR1 signalling prolongs ACA activation [?]; PKB→CRAC is a model assumption.
$k_{Cr,P}$	1	$\mu\text{M}^{-1}\text{s}^{-1}$	E	PIP3-facilitated recruitment of CRAC, which is a PH-domain protein [?], [?], [?].
$k_{Cr,b}$	0.2	s^{-1}	C	PIP3-independent membrane association; keeps the relay functional when PIP3 is absent (PI3K-independent PKB signalling [?]).
$k_{Cr,off}$	0.1	s^{-1}	E	Release of unphosphorylated CRAC from the membrane.
$k_{Cr,dp}$	1.5	s^{-1}	D	$= k_{Cr} a (1 - f)/f$ with $a = 567 \text{ PKBA}^*/\text{voxel}$ (the measured peak) and target CRAC* fraction $f = 0.44$; fixes the activated fraction of the CRAC switch.
k_X	200	$\text{molec s}^{-1} \text{ cell}^{-1}$	C	Injection rate of X at full CRAC* (per-CRAC* rate $k_X/N_{CRAC} = 200/2002 = 0.1 \text{ s}^{-1}$, with $N_{CRAC} = 2002$ per cell); must exceed the ≈ 24 -molecule threshold of Module 0.
K_P, n_P	5000, 8	$\text{molec/voxel}, -$	C	Half-open level and steepness of the PIP3 gate (Sec. ??).
k_{PKA}	0.938	$\mu\text{M}^{-1}\text{s}^{-1}$	D	PKA* feedback on PKBR1*; PKA exerts negative feedback on Ras, Rap1 and TORC2 signalling [?]. Derived from the Laub-Loomis coefficient 0.6 min^{-1} [?] ($30\times$ boosted), normalised by the PKA pool.
$[\text{RasC}]_{\text{tot}}$	385	molec/voxel	E	Pool size; not constrained by measurements.
$[\text{CGAP}]_{\text{tot}}, [\text{TORC2}]_{\text{tot}}, [\text{CRAC}]_{\text{tot}}$	154	molec/voxel	E	Pool sizes (2002 per cell); not constrained by measurements.
$[\text{PKBR1}]_{\text{tot}}$	77	molec/voxel	E	PKBR1 pool; PkbA (Module 4) is a separate, larger pool.
$D_{\text{RasC}}, D_{\text{TORC2}}, D_{\text{PKBR1}}, D_{\text{CRAC}_m}$	0.1	$\mu\text{m}^2/\text{s}$	E	Membrane-bound species.
$D_{\text{GEFA}}, D_{\text{CGAP}}, D_{\text{CRAC}_c}$	10	$\mu\text{m}^2/\text{s}$	E	Cytosolic value ($D_{\text{GEFA}}^* = 2$).

c) *Constants.*: Most pool sizes and the RasC-branch rate constants are estimates for which no measurements exist; the calibrated properties are the timing (TORC2* and PKBR1* peak within 5–10 s), the activated fraction of the CRAC switch and the relay drive. Two constants are derived from measurements of other modules and must be re-derived when those change. First, $k_{Cr,dp}$ sets the activated fraction of the CRAC phospho-cycle, which is a function of $a/k_{Cr,dp}$ only, where $a = k_{Cr}[\text{PKBA}^*]$; it is derived from the measured peak of PkbA* (567 molecules per voxel) and the target CRAC* fraction of 44 %. When Module 4 was re-anchored the measured peak fell tenfold, the fraction fell to 11 % and the relay stopped firing, so the derivation is stored with the measurement it depends on. Second, the half-open level K_P of the PIP3 gate is placed between the PIP3 peak evoked by 2 nM (about 5000 molecules per voxel) and the standing PIP3 (about 10^3). Neither the gain k_X nor k_{Cr} can replace these derivations: both would restore the product of drive and fraction but leave a switch operating in a tenth of its range, without its contrast. The pairing $k_{A,on} = k_{E,on}k_{A,off}/k_{E,off}$ removes a saturation that the earlier parameterisation had (RasGEFA 83 % active at rest and 99 % at 60 nM), where a cAMP step drove RasC-GTP down instead of up; RasC is activated rapidly and transiently by cAMP [?]. The pool $[\text{GEFA}]_{\text{tot}}$ was chosen so that the number of active GEFs at the driven state is preserved. The relay is nearly dose independent in the model: 2 nM cAMP fires the excitable element as fully as 200 nM, because the LEGI/PIP3 stage reports contrast rather than level; this is why the gate must be steep.

J. Module 9: cAMP Synthesis and Export

a) *Role.*: Cells lacking the aggregation-stage adenylyl cyclase ACA have almost no cyclase activity and fail to aggregate, whereas motility and chemotaxis are unaffected, so cAMP synthesis is what allows cell–cell communication [?]. ACA is activated in response to cAMP through CRAC [?] and the MAP kinase ERK2 [?]; after a sustained stimulus its activity peaks after about 30 s and returns to baseline with a half-time of 1.8 min [?]. cAMP is exported by ABC transporters, of which AbcB3 is a component [?], and the rate of secretion is proportional to the intracellular cAMP concentration [?]. Cells released about 5 pmol cAMP per 10^6 cells, i.e. about 3×10^6 molecules per cell, in response to a 2-min stimulus at 7 h of development [?].

b) *Reactions.*:

- 9.1/9.2 The active cyclase ACA* makes cAMP inside the cell. The synthesis is split into a Hunger-scaled arm and a constant arm, so that starved cells, which are closer to the oscillatory regime, make more cAMP per active cyclase; the two arms sum to 20 s^{-1} per ACA*.
- 9.2b cAMP is exported to the extracellular medium. The export is first order in the intracellular concentration [?] and is the only cAMP output of the cell: removing it removes the secreted signal although the intracellular cascade still fires.
- 9.3 PKA* inactivates ACA*, the negative feedback of the Laub–Loomis circuit that ends the pulse [?].
- 8.12 ACA* decays spontaneously. The rate reproduces the decline of ACA activity after a sustained stimulus [?]. At the much faster value used earlier, ACA* was slaved to the X spike and the pulse carried only

TABLE XVI

MODULE 9 REACTIONS: cAMP SYNTHESIS BY THE ADENYLYL CYCLASE ACA, EXPORT, AND ACA INACTIVATION. REACTION 8.12 BELONGS TO THE ACA ARM AND IS LISTED HERE.

ID	Reaction	Propensity (rate law)
9.1	$ACA^* \rightarrow ACA^* + cAMP_i$	$k_9 (1 + \alpha_H H/H_{\max})[ACA^*]$
9.2	$ACA^* \rightarrow ACA^* + cAMP_i$	$k_{11}[ACA^*]$
9.2b	$cAMP_i \rightarrow cAMP_e$	$k_x[cAMP_i]$
9.3	$PKA^* + ACA^* \rightarrow PKA^* + ACA$	$k'_2[PKA^*][ACA^*]$
8.12	$ACA^* \rightarrow ACA$	$k_{Ao}[ACA^*]$

TABLE XVII
MODULE 9 CONSTANTS.

Symbol	Value	Unit	Basis	Meaning, justification, source
k_9	8	s^{-1}	C	Hunger-scaled synthesis arm; $k_9 + k_{11} = 20 s^{-1}$ is an ordinary adenylyl-cyclase turnover chosen so that one pulse exports $\approx 9 \times 10^6$ cAMP molecules per cell, three times the 3×10^6 measured for a 2 min stimulus at 7 h [?].
k_{11}	12	s^{-1}	C	Constant synthesis arm (see k_9).
α_H	0.5		E	Hunger amplification of synthesis ($1 + \alpha_H H/H_{\max}$); starved cells make more cAMP per active ACA.
k_x	0.5	s^{-1}	C	First-order export via the ABC transporter AbcB3 [?]; the secretion rate is proportional to intracellular cAMP [?]. $\tau = 2$ s, short against the ≈ 40 s pulse, so export adds no delay; low enough that RegA (Module 10) controls the exported fraction.
k'_2	0.0469	$\mu M^{-1} s^{-1}$	D	PKA-dependent negative feedback on ACA; from the Laub–Loomis coefficient 0.9 min^{-1} [?] divided by the PKA pool ($0.32 \mu M$).
k_{Ao}	6.4×10^{-3}	s^{-1}	L	Intrinsic ACA^* decay, $t_{1/2} = 108$ s, the decline of ACA activity after a sustained stimulus (peak at 30 s, $t_{1/2} = 1.8 \text{ min}$ [?]); the realised decline is faster because 9.3 adds to it.
$[ACA]_{\text{tot}}$	865	molec/voxel	C	See Module 0.
D_{cAMP_i}	30	$\mu m^2/s$	E	Buffered effective cytosolic diffusion of cAMP.
D_{ACA}	0.1	$\mu m^2/s$	E	Membrane-bound cyclase.

8×10^5 molecules; at the present value the brief X spike charges ACA^* , and the pulse length is set by the $ACA^*/PKA/RegA$ loop and not by X.

c) *Constants.*: The amplitude is anchored to the measured relay output and not to the normalised synthesis coefficients of [?]: the pool of 865 ACA molecules per voxel and the turnover of $20 s^{-1}$ were chosen so that one pulse exports about 9×10^6 molecules per cell. This is three times the 3×10^6 molecules measured for a 2-min stimulus [?], so the modelled pulse is at the upper end of measured amplitudes. The cyclase pool has never been measured. The export rate k_x was lowered from an earlier derived value of $3 s^{-1}$: at that value export outran the phosphodiesterase RegA by a factor of 250, and knocking out the whole $ERK2 \rightarrow RegA$ edge changed the exported cAMP by a factor of 1.0. At $0.5 s^{-1}$ the exported fraction is 0.05 at rest (RegA armed) and 0.83 at the pulse peak (RegA disarmed), so the contrast between rest and pulse is carried by RegA and not by the transporter. *Limitation*: only a first-order dependence of export on $cAMP_i$ is established [?]; its numerical value is a calibration.

K. Module 10: RegA, PKA and ERK2 Brake

a) *Role.*: The intracellular phosphodiesterase RegA (DdPDE2) degrades intracellular cAMP [?]. It carries a phospho-relay receiver domain: phosphorylation of Asp212 by the histidine phospho-transfer protein RdeA activates the enzyme at least 20-fold [?], and its cAMP-hydrolysing domain has a Michaelis constant of about $5 \mu M$ [?]. Loss of

RegA activates PKA and disturbs differentiation [?]. cAMP oscillations in populations are generated by a circuit in which cAMP activates PKA, PKA inhibits ERK2, and ERK2 controls cAMP levels through RegA; the ERK2 activation oscillates in phase with the cAMP peaks [?], [?]. PKA also shapes ERK2 activation and adaptation [?]. Module 10 is the resulting brake: at rest RegA holds intracellular cAMP near zero, an ERK2 spike disarms it so that cAMP can accumulate, and its recovery and the PKA arm terminate the pulse and set the refractory tail.

b) Reactions.

- 10.B1/10.B2 The phospho-relay activates RegA and reverses. The pair sets the resting active fraction, 0.9. Only the lumped transfer from RdeA to RegA is modelled [?].
- 10.B3 $ERK2^*$ converts active RegA into an inactive, ERK2-modified form. The model implements the control of RegA by ERK2 [?] as inactivation, the sign required for an ERK2 spike to escape the brake.
- 10.B4 The inactive form returns to the active form ($\tau = 60$ s). This reset step is the one that terminates each pulse; without it RegA would be lost irreversibly.
- 10.C1/10.C2 RegA hydrolyses $cAMP_i$. The rate is linear in cAMP because $cAMP_i$ stays far below the Michaelis constant; the unphosphorylated form keeps 1/20 of the activity of the phosphorylated form [?].
- 10.D1/10.D2 cAMP activates PKA by binding to its regulatory subunit, and PKA^* decays with $\tau = 40$ s, which sets the minutes-long tail of the refractory period.

TABLE XVIII

MODULE 10 REACTIONS: THE INTRACELLULAR BRAKE. REGA IS A CONSERVED THREE-STATE MOIETY: REGA_u (UNPHOSPHORYLATED, WEAKLY ACTIVE), REGA_p (PHOSPHORYLATED AT D212, ACTIVE) AND REGA_{pi} (PHOSPHORYLATED BY ERK2, INACTIVE). PKA AND ERK2 ARE THE KINASES OF THE LAUB–LOOMIS/MAEDA CIRCUIT.

ID	Reaction	Propensity (rate law)
10.B1	RegA _u → RegA _p	$k_{B1}[\text{RegA}_u]$
10.B2	RegA _p → RegA _u	$k_{B2}[\text{RegA}_p]$
10.B3	RegA _p + ERK2* → RegA _{pi} + ERK2*	$k_{B3}[\text{RegA}_p][\text{ERK2}^*]$
10.B4	RegA _{pi} → RegA _p	$k_{B4}[\text{RegA}_{pi}]$
10.C1	cAMP _i + RegA _p → RegA _p	$k_{\text{cat}}[\text{cAMP}_i][\text{RegA}_p]$
10.C2	cAMP _i + RegA _u → RegA _u	$\phi k_{\text{cat}}[\text{cAMP}_i][\text{RegA}_u]$
10.D1	cAMP _i + PKA → cAMP _i + PKA*	$k_{D1}[\text{cAMP}_i][\text{PKA}]$
10.D2	PKA* → PKA	$k_{D2}[\text{PKA}^*]$
10.D3	PKA* + ERK2* → PKA* + ERK2	$k_{D3}[\text{PKA}^*][\text{ERK2}^*]$

TABLE XIX
MODULE 10 CONSTANTS.

Symbol	Value	Unit	Basis	Meaning, justification, source
[RegA] _{tot}	1540	molec/voxel	E	Copy number never measured; 0.32 μM puts RegA on the scale of ERK2 and PKA and keeps the required catalytic efficiency sub-diffusion-limited.
[PKA] _{tot}	1540	molec/voxel	E	0.32 μM.
k_{B1}	0.05	s ⁻¹	E	Lumped phospho-relay (RdeA→RegA D212) that activates RegA [?], [?].
k_{B2}	5.6×10^{-3}	s ⁻¹	C	Reverse phospho-transfer [?]; chosen for a resting active fraction of 0.9.
k_{B3}	8	μM ⁻¹ s ⁻¹	C	ERK2* inactivates RegA [?], [?]. The Laub–Loomis coefficient is a normalised-activity rate, not a molecular one; the value gives $\tau \approx 0.5$ s at the ERK2* peak so that the brake is released inside the X spike.
k_{B4}	0.0167	s ⁻¹	D	The reset reaction ($\tau = 60$ s) that ends each pulse; taken from the Laub–Loomis reset coefficient [?].
k_{cat}	30	μM ⁻¹ s ⁻¹	D	$k_{\text{cat}}/K_M = 3 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$; $k_{\text{cat}} \approx 150 \text{ s}^{-1}$ at the measured $K_M \approx 5 \text{ μM}$ [?].
ϕ	0.05		L	Activity of RegA _u relative to RegA _p ; phosphorylation activates RegA by at least 20-fold [?].
k_{D1}	0.25	μM ⁻¹ s ⁻¹	E	cAMP binding to the PKA regulatory subunit; with k_{D2} this gives $K_d = k_{D2}/k_{D1} = 0.1 \text{ μM}$ (model choice; PKA activation is the second element of the circuit of [?]).
k_{D2}	0.025	s ⁻¹	D	PKA* decay ($\tau = 40$ s), from the Laub–Loomis coefficient 1.5 min^{-1} [?]; sets the minutes-long refractory tail.
k_{D3}	0.0417	μM ⁻¹ s ⁻¹	D	PKA* inhibits ERK2, from the coefficient 0.8 min^{-1} [?]; PKA also shapes ERK2 activation and adaptation [?].
$D_{\text{RegA}}, D_{\text{PKA}}, D_{\text{ERK2}}$	10	μm ² /s	E	Cytosolic value.

10.D3 PKA* inactivates ERK2*, which closes the loop [?], [?].

c) *Constants.*: The Laub–Loomis coefficients are rates of a normalised-activity model and not molecular rate constants, so they cannot be used literally. Used literally as a bimolecular rate constant ($1.3 \text{ min}^{-1} = 0.022 \text{ μM}^{-1}\text{s}^{-1}$), the coefficient for RegA inactivation would disarm RegA with $\tau = 144$ s even with the whole ERK2 pool active, while ERK2* is up for only about 45 s, and the brake would never open (knock-down 1.3×); the present value gives a knock-down of RegA_p of 64×. The cyclase-RegA competition needs $k_{\text{cat}}/K_M \approx 3 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$ with the RegA pool of 0.32 μM, which is sub-diffusion-limited: the pool was enlarged from an earlier 0.032 μM, which was itself a normalised-unit value and required a catalytic efficiency near the diffusion limit before RegA could compete with export. The copy numbers of RegA, PKA and ERK2 have not been measured. The coefficients that are taken from the Laub–Loomis parameter set (k_{B4} , k_{D2} , k_{D3} , and k'_2 of Module 9) are converted to molecular rates by dividing by the pool sizes and are calibrated only in the sense

that the resulting pulse (an export of about 10^7 molecules, a RegA knock-down of 64× and a full reset within about 6 min) is reproduced.

L. Module 11: Extracellular Phosphodiesterase PdsA and Inhibitor PdiA

a) *Role.*: The level of extracellular cAMP is controlled by a phosphodiesterase that is secreted by the cells [?]; extracellular cAMP is degraded predominantly by DdPDE1 (PdsA) and its close homologue [?]. Cells lacking PdsA respond chemotactically to a narrower range of cAMP concentrations and do not form streams at low density [?], and PdsA-null cells cannot move in a coordinated manner in mounds [?]. The *pdsA* gene has separate promoters for growth, aggregation and late development [?], and the aggregative promoter is switched off after aggregation [?]. PdsA is inhibited by a secreted glycoprotein, PdiA, with 1:1 stoichiometry [?]; the inhibitor is absent in growing cells and appears after transfer to starvation buffer [?], and its level is regulated independently by exogenous cAMP [?]. Module 11 therefore contains the

TABLE XX

MODULE 11 REACTIONS: THE EXTRACELLULAR PHOSPHODIESTERASE PdsA AND ITS INHIBITOR PdiA. PdsA_i, PdsA_m AND PdsA_e ARE THE INTRACELLULAR PRECURSOR, THE MEMBRANE-DISPLAYED AND THE SECRETED ENZYME; cAMP_s IS A SLOW, LOW-PASS FILTERED COPY OF THE LOCAL cAMP_e. REACTIONS MARKED * RUN TWICE: INSIDE CELLS AND, WITH IDENTICAL CONSTANTS, ON AGENT-FREE VOXELS OF THE WORLD. $P(H) = H^2/(H_p^2 + H^2)$ AND $r(\text{cAMP}_s) = [1 + (\text{cAMP}_s/K_{\text{pdi}})^2]^{-1}$ ARE THE HILL GATES OF SEC. ??.

ID	Reaction	Propensity (rate law)
11.D1a	$\emptyset \rightarrow \text{PdsA}_i$	k_{bas} (zeroth order)
11.D1b	$\emptyset \rightarrow \text{PdsA}_i$	$k_{\text{ind}} P(H)$ (zeroth order)
11.D2	$\text{PdsA}_i \rightarrow \text{PdsA}_m$	$k_{\text{sec}}[\text{PdsA}_i]$
11.D3	$\text{PdsA}_m \rightarrow \text{PdsA}_e$	$k_{\text{shed}}[\text{PdsA}_m]$
11.D4a,b	$\text{PdsA}_m \rightarrow \emptyset$; $\text{PdsA}_i \rightarrow \emptyset$	$k_{\text{deg}}[\cdot]$
11.D4c*	$\text{PdsA}_e \rightarrow \emptyset$	$k_{\text{deg}}[\text{PdsA}_e]$
11.D4d*	$\text{PdsA:PdiA} \rightarrow \emptyset$	$k_{\text{deg}}[\text{PdsA:PdiA}]$
11.F1	$\text{cAMP}_e \rightarrow \text{cAMP}_e + \text{cAMP}_s$	$k_{\text{trk}}[\text{cAMP}_e]$
11.F2	$\text{cAMP}_s \rightarrow \emptyset$	$k_{\text{trk}}[\text{cAMP}_s]$
11.E1	$\emptyset \rightarrow \text{PdiA}$	$k_{\text{pdi}} P(H) r(\text{cAMP}_s)$ (zeroth order)
11.E2*	$\text{PdiA} \rightarrow \emptyset$	$k_{\text{deg}}[\text{PdiA}]$
11.E3f*	$\text{PdsA}_e + \text{PdiA} \rightarrow \text{PdsA:PdiA}$	$k_{i,\text{on}}[\text{PdsA}_e][\text{PdiA}]$
11.E3r*	$\text{PdsA:PdiA} \rightarrow \text{PdsA}_e + \text{PdiA}$	$k_{i,\text{off}}[\text{PdsA:PdiA}]$
11.E4*	$\text{cAMP}_e + \text{PdsA:PdiA} \rightarrow \text{PdsA:PdiA}$	$k_{\text{res}}[\text{cAMP}_e][\text{PdsA:PdiA}]$
11.2*	$\text{cAMP}_e + \text{PdsA}_e \rightarrow \text{PdsA}_e$	$k_{\text{pdsa}}[\text{cAMP}_e][\text{PdsA}_e]$

synthesis and secretion chain of the enzyme, a starvation-induced inhibitor that is repressed by the cAMP history, and their reversible binding.

b) Reactions.:

- 11.D1a PdsA is synthesised in the cell at a constitutive rate (11.D1a) and at a starvation-induced rate gated by $P(H)$ (11.D1b), so that the enzyme appears with development. The gate depends only on the developmental variable H ; the model contains no dependence of PdsA synthesis on pulses of cAMP.
- 11.D2/D3 The enzyme is transferred from an intracellular precursor to the cell surface (11.D2) and shed into the medium (11.D3) as PdsA_e, which diffuses. Only the secreted form clears extracellular cAMP; the surface-bound form is an explicit, non-diffusing species.
- 11.D4 All protein states turn over with one rate constant. The reactions for the secreted enzyme and the complex run in cells and on free voxels, so that clearance takes place everywhere.
- 11.F1/F2 A per-voxel low-pass filter of cAMP_e, cAMP_s, lets PdiA repression follow the envelope of the pulse train and not individual pulses. Its time constant of 42 min is a modelling choice.
- 11.E1/E2 PdiA is synthesised under the same starvation gate as PdsA and repressed by cAMP_s [?], and degraded like PdsA.
- 11.E3 PdsA and PdiA bind reversibly with 1:1 stoichiometry [?]. The inhibition is titratable: as cells pack, the total pools scale with density, and the fraction of enzyme that is complexed rises.
- 11.E4 The complex retains a small residual activity, 10^{-3} of the free enzyme.
- 11.2 Free PdsA hydrolyses cAMP_e (the enzyme is a catalyst and is not consumed); this is the primary clearance of the extracellular signal.

c) *Constants and the signal range.*: Because PdsA_e is produced per cell and lost per voxel, its steady-state level, and hence the clearance strength $k_{\text{eff}} = k_{\text{pdsa}}[\text{PdsA}_e]$, is proportional to the cell density ρ . The range over which a pulse signals, $\lambda = \sqrt{D_{\text{cAMP}}/k_{\text{eff}}} \propto \rho^{-1/2}$, scales like the mean cell spacing, so that λ in units of cell spacings is independent of density for fixed rates, and the inhibitor titration breaks this only in the harmful direction: the fraction of complexed enzyme rises from 4% at $\rho = 7.7 \times 10^{-4}$ to about 70% at 6×10^{-2} cells per voxel, and λ in spacings grows as cells aggregate. The synthesis constants k_{bas} and k_{ind} were raised by a factor of 16 above the values that reproduce a $2 \mu\text{M}$ enzyme field at one cell per 36 voxels. With the smaller values the clearance constant was $k_{\text{eff}} = 9.6 \times 10^{-4} \text{ s}^{-1}$, so that $\lambda \approx 600 \mu\text{m}$ covered the whole population and the population fired as one synchronous global flash; at the present values $k_{\text{eff}} = 1.5 \times 10^{-2} \text{ s}^{-1}$ (clearance time 65 s) and $\lambda \approx 150 \mu\text{m}$ for a mean spacing of $50 \mu\text{m}$, i.e. about 1–3 cell spacings depending on the domain. The enzyme field is initialised with its steady-state profile and not with a uniform value: the world boundary is absorbing, and $\sqrt{D_{\text{PdsA}}/k_{\text{deg}}} = 361 \mu\text{m}$ is comparable to the domain, so that the boundary and not the degradation is the dominant sink, and the sustained field is 4.4–4.7 times higher at the centre than at the edge of the cell region. The steady state depends on the geometry, so λ has to be quoted per geometry.

Two constants deviate from measurements and should be read as model choices. The measured dissociation constant of the PdsA–PdiA complex is about 10^{-10} M [?]; the model uses $1 \mu\text{M}$, so that the binding equilibrates within seconds and clearance is not delayed. And the effective clearance constant, $1.2 \times 10^5 \text{ M}^{-1} \text{ s}^{-1}$, is derived from a normalised coefficient [?] and is not a measured k_{cat}/K_M ; the model compensates with a high enzyme concentration, and the surface pool of PdsA is 2.5×10^6 molecules per cell, which is a consequence of the slow shedding rate and the required flux and not a measured

TABLE XXI
MODULE 11 CONSTANTS. $M = 4817.6$ MOLECULES μM^{-1} PER VOXEL.

Symbol	Value	Unit	Basis	Meaning, justification, source
k_{pdsa}	0.122	$\mu\text{M}^{-1}\text{s}^{-1}$	D	Effective second-order clearance constant, $1.5 \times 4.9 \text{ min}^{-1}$: the external-cAMP degradation coefficient of [?], scaled by 1.5. Extracellular cAMP is degraded mainly by the secreted DdPDE1 (PdsA) [?].
$k_{\text{bas}}, k_{\text{ind}}$	$4.48 \times 10^{-3}, 4 \times 10^{-2}$	$\mu\text{M s}^{-1} \text{ voxel}^{-1}$	C	Constitutive and starvation-induced PdsA synthesis; $16\times$ the values that reproduce a $2 \mu\text{M}$ enzyme field at 1 cell per 36 voxels, so that the signal range λ is $\approx 1\text{--}3$ cell spacings and pulses are not global flashes (Sec. ??).
H_P, n_H	25, 2	molec/voxel, —	E	Half-induction level and steepness of $P(H)$; the window opens early relative to the Hunger ramp. The <i>pdsA</i> gene has an aggregation-specific promoter [?] that is switched off after aggregation [?]; that later closing is outside the simulated time span.
k_{sec}	0.01	s^{-1}	E	Precursor-to-surface transfer ($\tau = 100 \text{ s}$).
k_{shed}	5×10^{-4}	s^{-1}	E	Shedding of surface enzyme into the medium; equal to k_{deg} , i.e. a 50:50 partition between shedding and loss.
k_{deg}	5×10^{-4}	s^{-1}	E	Turnover of all PdsA/PdiA protein states ($\tau = 2000 \text{ s}$). This single constant sets the enzyme decay length $\sqrt{D/k_{\text{deg}}} = 361 \mu\text{m}$.
k_{pdi}	2.5×10^{-3}	$\mu\text{M s}^{-1} \text{ voxel}^{-1}$	E	Induced PdiA synthesis, co-induced with PdsA by the same starvation gate; PdiA is absent in growing cells and appears after transfer to starvation buffer [?].
$K_{\text{pdi}}, n_{\text{pdi}}$	0.05, 2	μM , —	E	Half-repression of PdiA synthesis by the slow cAMP signal cAMP_s [?].
k_{trk}	4×10^{-4}	s^{-1}	E	Unity-gain low-pass filter ($\tau = 42 \text{ min}$) of cAMP_e so that PdiA repression follows the pulse envelope.
$k_{i,\text{on}}, k_{i,\text{off}}$	1, 1	$\mu\text{M}^{-1}\text{s}^{-1}, \text{s}^{-1}$	E	1:1 binding with $K_d = 1 \mu\text{M}$. The measured stoichiometry is 1:1, but the measured K_d is $\approx 10^{-10} \text{ M}$ [?]; the model's K_d is 10^4 times weaker to keep the equilibration fast against clearance.
k_{res}	1.22×10^{-4}	$\mu\text{M}^{-1}\text{s}^{-1}$	E	Residual activity of the complex, $k_{\text{pdsa}}/1000$.
D_{cAMP_e}	350	$\mu\text{m}^2/\text{s}$	D	Free-solution value $444 \mu\text{m}^2/\text{s}$ ($4.44 \times 10^{-6} \text{ cm}^2/\text{s}$) [?] reduced by a tortuosity factor; cells are not excluded volumes on the lattice.
$D_{\text{PdsA}_e}, D_{\text{PdsA:PdiA}}$	65	$\mu\text{m}^2/\text{s}$	E	Secreted enzyme (Diffusion set A of the model description).
D_{PdiA}	60	$\mu\text{m}^2/\text{s}$	E	Soluble inhibitor.
$D_{\text{PdsA}_i}, D_{\text{PdsA}_m}, D_{\text{cAMP}_s}$	10, 0.1, 0	$\mu\text{m}^2/\text{s}$	E	Precursor (cytosolic), surface form (membrane) and the local filter state (non-diffusing).

abundance. The extracellular diffusion constant of cAMP is reduced from the free-solution value of $4.44 \times 10^{-6} \text{ cm}^2/\text{s}$ [?] to $350 \mu\text{m}^2/\text{s}$, because cells are not excluded volumes on the lattice.

the erasure time of 3 min is chosen to bridge the back of a wave.

M. Optional Module: Directional Memory

This module is off in the default network and is described for completeness. Cells keep their direction of motion in the back of a wave for the natural period of 6 min, but increasingly reverse for longer periods; the molecular carrier of this memory is not known [?]. The optional module represents it phenomenologically by a membrane mark that is written where PIP3 is high (M.1) and erased slowly (M.2), and that does not feed back onto any other module; the motility reads it, so that cells steer on where PIP3 has been over the last minutes and not on its instantaneous value. Because the mark is the time integral of the locally gated PIP3 with leakage $\tau = 1/k_e$, its centroid averages the PIP3 direction over the writing window and holds it after PIP3 falls. No verified molecule is both PIP3-driven and long-lived, so the pool, the writing rate and the diffusion constant are estimates (Table ??);

III. SECTION 2

IV. SECTION 3

V. CONCLUSION

TABLE XXII

OPTIONAL MEMORY MODULE (MEM), OFF BY DEFAULT AND NOT USED FOR THE RESULTS OF THIS PAPER: A PHENOMENOLOGICAL MARK WRITTEN WHERE PIP3 IS HIGH AND ERASED SLOWLY. $H(\text{PIP3}) = N_{\text{PIP3}}^3 / (K_{\text{mem}}^3 + N_{\text{PIP3}}^3)$.

ID	Reaction	Propensity (rate law)
M.1	$\text{Mem}_c \rightarrow \text{Mem}_m^*$	$k_w [\text{Mem}_c] H(\text{PIP3})$
M.2	$\text{Mem}_m^* \rightarrow \text{Mem}_c$	$k_e [\text{Mem}_m^*]$

TABLE XXIII

CONSTANTS OF THE OPTIONAL MEMORY MODULE.

Symbol	Value	Unit	Basis	Meaning, justification, source
$[\text{Mem}]_{\text{tot}}$	5000	molec/voxel	E	Large pool so that the centroid of the mark is not shot-noise limited.
k_w	4×10^{-3}	s^{-1}	E	Writing rate, chosen so that less than $\approx 40\%$ of the pool is written per wave (an unsaturated integrator).
$K_{\text{mem}}, n_{\text{mem}}$	5000, 3	molec/voxel, —	C	PIP3 level of half-writing, above resting PIP3 and inside the range reached on the rising edge of a wave.
k_e	5.56×10^{-3}	s^{-1}	E	Erasure with $\tau = 3$ min, so that the mark persists through the back of a wave: cells keep their direction there for waves of the natural 6 min period [?].
$D_{\text{Mem}_c}, D_{\text{Mem}_m}$	10, 0.01	$\mu\text{m}^2/\text{s}$	E	Cytosolic carrier and nearly immobile mark.